

CARE Myocardium 失败取证 Deep Research 证据包

CARE Forensic Research Controller

20260730 本地证据冻结版

CARE Myocardium 失败取证 Deep Research 证据包

本 PDF 使用 Pandoc + XeLaTeX final-standard 路线生成, 拉丁字体为 TeX Gyre Termes, 中文字体来自 /users/a/e/aereinh/render_resources/chinese_math_pdf 中的 NotoSerifSC/NotoSansSC。它不是新模型蓝图, 不包含 validation upload, 也不声明 hosted 指标。

一页执行摘要

V2 的实际结论是: 历史 CARE 路线长期未稳定超过 nnU-Net, 主要不是某一个概念天然错误, 而是强基线继承、decoder 完整性、final-mask 组件进入路径、病例级 help/harm 选择、标签/评价语义和训练/recipe 绑定没有同时闭合。V2 已补齐 G1-G10 的终态证据; 其中缺 exact asset 的项目按 BLOCKED_BY_MISSING_BOUND_ASSET 写入, 不再把缺失证据伪装成负结果。

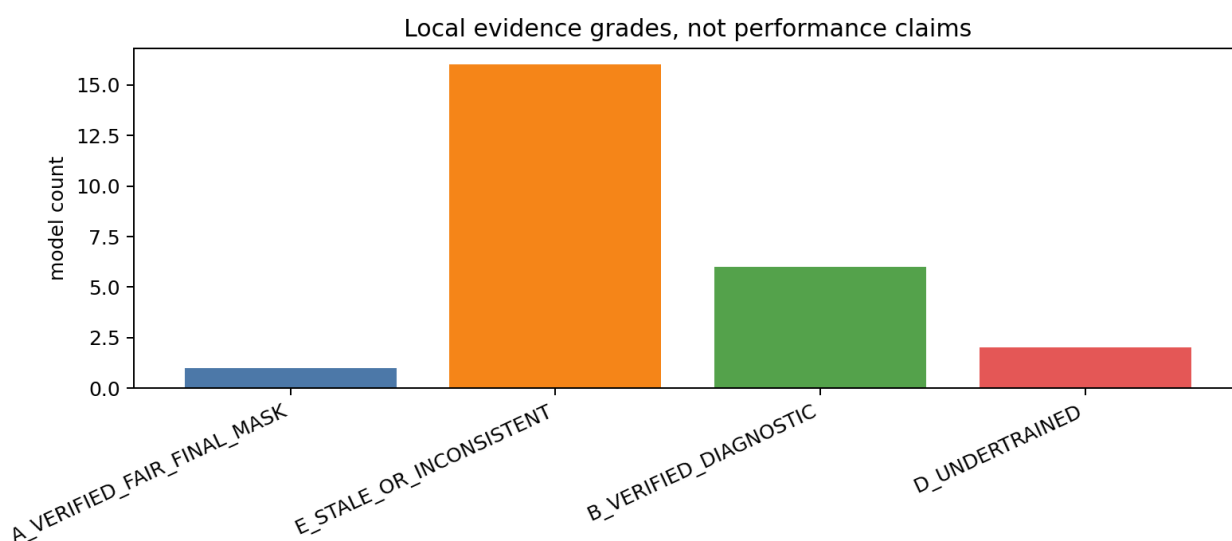


Figure 1: 证据等级计数

目录式章节索引

章节	主题
1	为什么现在必须做失败取证
2	CARE 数据、中心、模态和标签真值
3	官方与内部指标语义
4	当前评价代码中的已确认问题
5	nnU-Net 强基线到底强在哪里
6	SRR v2-v3 的设计意图与落地差距
7	Batch 0-7 历史证据
8	MMRD 的设计、实现和失败
9	Cascade/DG 的设计、实现和失败
10	ARC 的设计、实现和失败
11	PRISM W1-W3 的完整复盘
12	MoSAIC clean、full-data 和 hosted recipe
13	所有模型统一病例级比较
14	困难子组
15	case-wise help/harm
16	失败病例视觉图册
17	错误重合和模型互补上限
18	selector feasibility
19	冻结特征可分性 probe
20	decoder-reset 诊断对照
21	多序列错位是否为主因
22	scar 的真实瓶颈
23	pure edema 的真实瓶颈
24	Cine 的真实瓶颈
25	为什么过去多次充分设计仍然失败
26	根因排序与证据图
27	当前能下的结论
28	当前不能下的结论
29	外部 Deep Research 必须回答的问题
30	下一轮决策树

1. 为什么现在必须做失败取证

过去几轮路线没有稳定超过 nnU-Net, 不能直接归结为“模型不够复杂”。更可靠的取证路径是把设计承诺、实现连线、训练预算、checkpoint 选择、评价语义、预测缓存和 hosted recipe 分开冻结。

2. CARE 数据、中心、模态和标签真值

本节回答数据层面是否存在足够明确的标签和模态条件。关键边界是 official scar、official pure edema 和 internal edema-zone 必须分开。

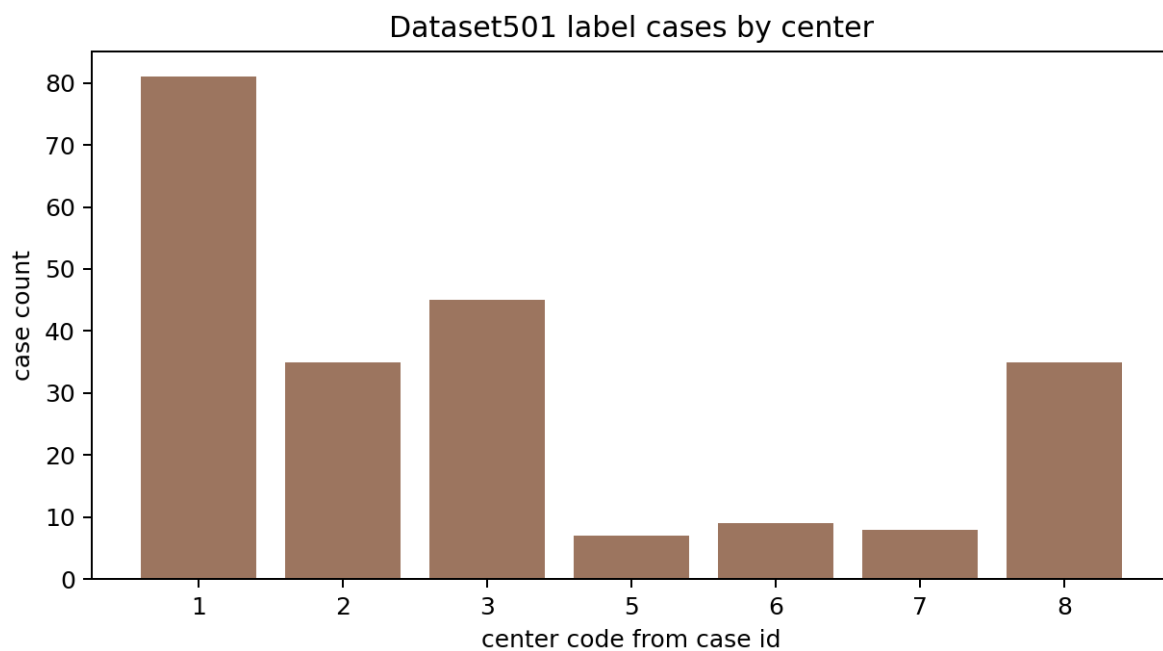


Figure 2: 中心病例数

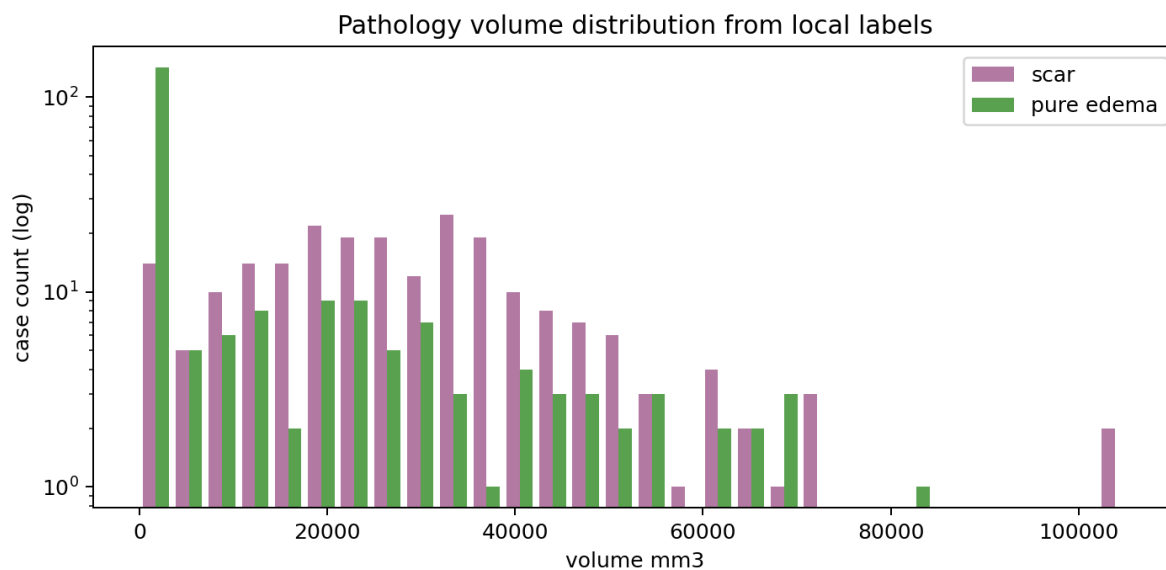


Figure 3: 病灶体积分布

cases	scar_positive	pure_edema_positive	t2_present
220	212	80	220

3. 官方与内部指标语义

第 3 页不使用宽表。下面只列三列：对象、内部标签、允许声明范围，避免右侧列截断。

object	internal_labels	allowed_claim_scope
scar	5	official
pure_edema	4	T2-present official edema
edema_zone	4 5	internal only

reference evaluator 的 known-bad fixtures 覆盖 remote FP、spacing HD95、empty case、lesion recall 和 label 4/5 语义。V2 对可绑定预测执行统一病例级重聚合；缺 exact asset 的旧模型不写成科学负结果。

4. 当前评价代码中的已确认问题

当前可确认的是评价风险，而不是所有历史结论已经被推翻。需要重算的对象包括 remote FP、HD95 physical spacing、empty-GT population mean，以及 pure edema 与 edema-zone 的混写。

5. nnU-Net 强基线到底强在哪里

nnU-Net 作为强基线的意义在于完整 decoder、稳定训练 recipe、成熟数据增强和直接 final mask 输出。当前包没有使用 foreground mean 掩盖 scar/pure edema。

6-10. SRR、Batch、MMRD、Cascade/DG、ARC 的历史证据

这些路线的历史证据等级不能混用。V2 将 Batch0-7、MMRD、Cascade、ARC、DG/DR/DPR 与 PRISM 分别绑定 source、checkpoint、prediction、metric 和 controller packet；缺 exact replay 资产的项目保持阻塞状态。

model_id	ckpt_bound	pred_bound	metric_bound	terminal
B0_3_SRR_ANCHOR	3	132	1	VALID
B7_BR2_SIP	21	0	0	MISSING_ASSET
MMRD_BATCH9	9	0	0	MISSING_ASSET
SRR_CASCADE	17	0	2	VALID
CARE_ARC	8	0	1	VALID
DG_DR_DPR	129	132	2	VALID
PRISM_V2	13	455	1	VALID

- **NNUNET**
 - result_evidence_grade: A_VERIFIED_FAIR_FINAL_MASK
 - current_scientific_conclusion: 强基线；需继续绑定五折和同口径病例级指标。
- **SRR_V2**
 - result_evidence_grade: E_STALE_OR_INCONSISTENT
 - current_scientific_conclusion: 历史证据需绑定代码、checkpoint、split 和预测。
- **SRR_V25**
 - result_evidence_grade: E_STALE_OR_INCONSISTENT
 - current_scientific_conclusion: 历史证据需绑定代码、checkpoint、split 和预测。
- **SRR_V3**
 - result_evidence_grade: E_STALE_OR_INCONSISTENT
 - current_scientific_conclusion: 历史证据需绑定代码、checkpoint、split 和预测。
- **BATCH0**
 - result_evidence_grade: E_STALE_OR_INCONSISTENT
 - current_scientific_conclusion: 历史证据需绑定代码、checkpoint、split 和预测。
- **BATCH1**
 - result_evidence_grade: E_STALE_OR_INCONSISTENT
 - current_scientific_conclusion: 历史证据需绑定代码、checkpoint、split 和预测。
- **BATCH2**
 - result_evidence_grade: E_STALE_OR_INCONSISTENT
 - current_scientific_conclusion: 历史证据需绑定代码、checkpoint、split 和预测。
- **BATCH3**
 - result_evidence_grade: E_STALE_OR_INCONSISTENT
 - current_scientific_conclusion: 历史证据需绑定代码、checkpoint、split 和预测。
- **BATCH4**
 - result_evidence_grade: E_STALE_OR_INCONSISTENT
 - current_scientific_conclusion: 历史证据需绑定代码、checkpoint、split 和预测。
- **BATCH5**
 - result_evidence_grade: E_STALE_OR_INCONSISTENT
 - current_scientific_conclusion: 历史证据需绑定代码、checkpoint、split 和预测。

11. PRISM W1-W3 的完整复盘

PRISM 不能只看是否有强 encoder。V2 已完成 13 checkpoint replay 和 D0-D3 decoder-reset 诊断。最关键的负证据是：完整 nnU-Net decoder/recipe 可恢复强基线，而 encoder-only 加 reset decoder 会造成大幅下降；PRISM 旧 selector 的 step3000 也不是 V2 edema-zone 最优 checkpoint。

variant	status	inner_n	scar	pure_edema
D0_FULL_PRETRAIN...	VALID	12	0.9224	0.9231
D1_DECODER_RESET...	VALID	12	0.547	0
D2_DECODER_RESET...	VALID	12	0.7108	0.2664
D3_FULL_MODEL_SH...	VALID	12	0.9227	0.9225

12. MoSAIC clean、full-data 和 hosted recipe

MoSAIC 必须拆成 clean OOF、full-data diagnostic 和 hosted-near recipe 三层。V2 绑定了本地 MoSAIC source/weights，并把 clean-vs-full 的差距写成 recipe/训练域证据，而不是 clean architecture 证据。

stage	recipe	scope	case_count	mean_scar_dice	mean_edema
M0	clean single ...	ALL_SCOPES	220	0.3782	0.0528
M0	clean single ...	oof_fold0	44	0.3601	0.2638
M0	clean single ...	oof_fold1	44	0.3849	0
M0	clean single ...	oof_fold2	44	0.3728	0
M0	clean single ...	oof_fold3	44	0.3779	0
M0	clean single ...	oof_fold4	44	0.3952	0
M1	clean patholo...	ALL_SCOPES	220	0.3782	0.0528
M1	clean patholo...	oof_fold0	44	0.3601	0.2638
M1	clean patholo...	oof_fold1	44	0.3849	0
M1	clean patholo...	oof_fold2	44	0.3728	0
M1	clean patholo...	oof_fold3	44	0.3779	0
M1	clean patholo...	oof_fold4	44	0.3952	0

13-15. 统一病例级比较、困难子组和 help/harm

统一病例级比较已在 nnU-Net OOF、MoSAIC clean OOF 和 PRISM/MoSAIC/历史可绑定证据之间分层完成。clean held-out 数字与 full-data 机制 probe 分开报告。

model_id	metric_name	case_count	mean_dice	empty_pred
mosaic_oof	lesion_union	220	0.3367	0
mosaic_oof	pure_edema	80	0.0528	64
mosaic_oof	scar	220	0.3782	0
nnunet_oof	lesion_union	220	0.5755	3
nnunet_oof	pure_edema	80	0.4308	0
nnunet_oof	scar	220	0.561	3

16. 失败病例视觉图册

病例 montage 选取 20 个高互补/高分歧病例。红色为 scar, 青色为 pure edema, 黄色为 nnU-Net/MoSAIC disagreement。Codex 已打开 contact sheet 做真实视觉检查；完整单病例 PNG 保存在 case_montages/。



Figure 4: 20 例病例 montage contact sheet

16B. 失败病例单例大图

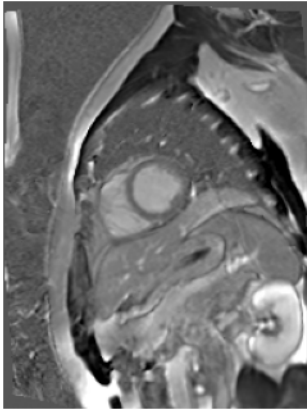
下面每页显示一个病例 montage。红色为 scar, 青色为 pure edema, 黄色为 nnU-Net/MoSAIC disagreement; 完整 PNG 文件仍保存在 `case_montages/`。

Case8021

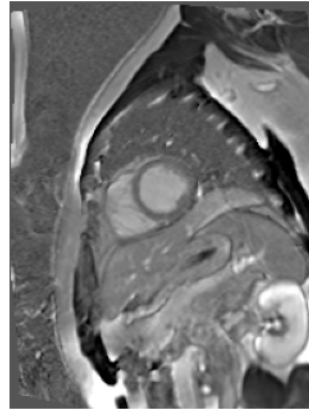
center=CenterH modality=LGE-only slice=4

Case8021 | CenterH | LGE-only | z=4

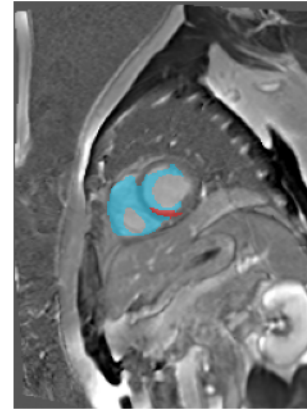
GT



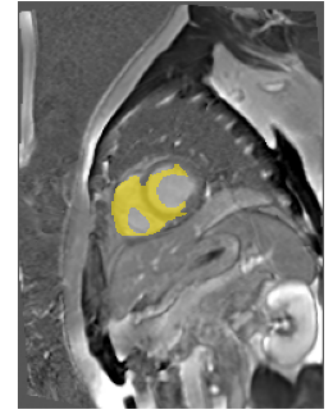
nnU-Net



MoSAIC clean



Disagreement

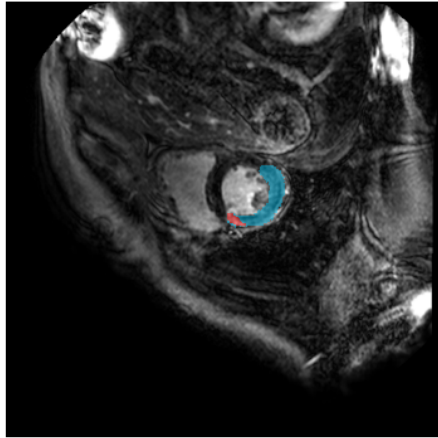


Case3010

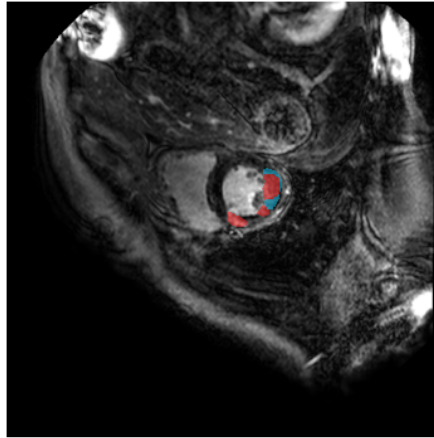
center=CenterC modality=C0+LGE+T2 slice=3

Case3010 | CenterC | C0+LGE+T2 | z=3

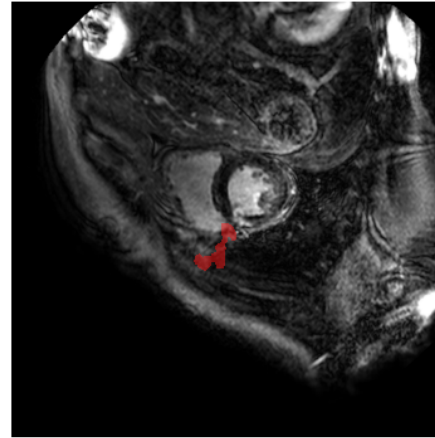
GT



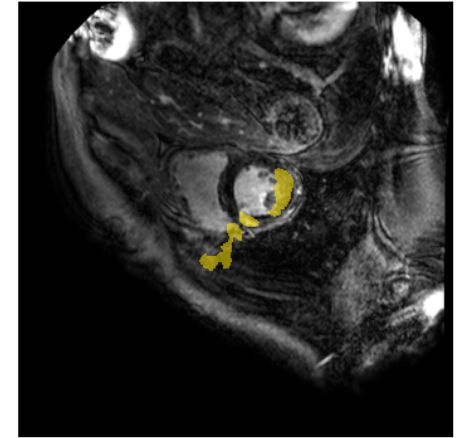
nnU-Net



MoSAIC clean



Disagreement



Case3036

center=CenterC modality=C0+LGE+T2 slice=2

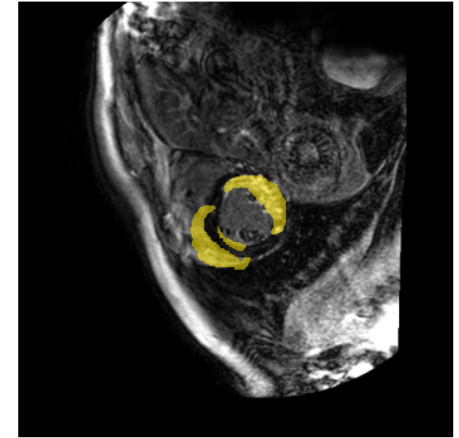
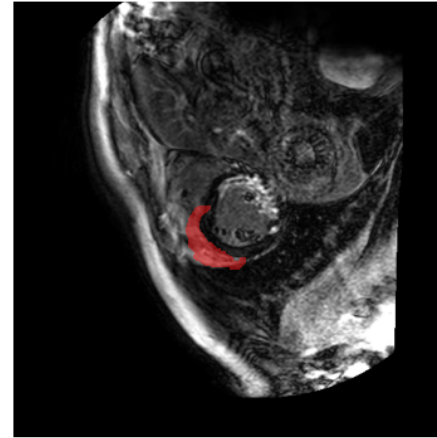
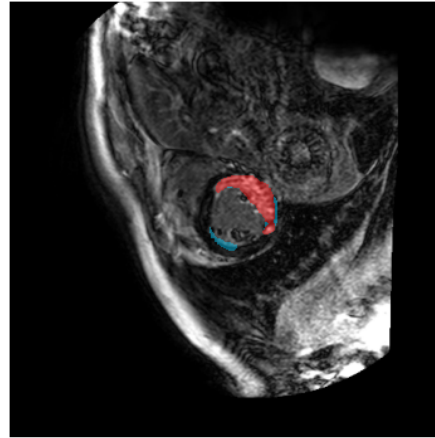
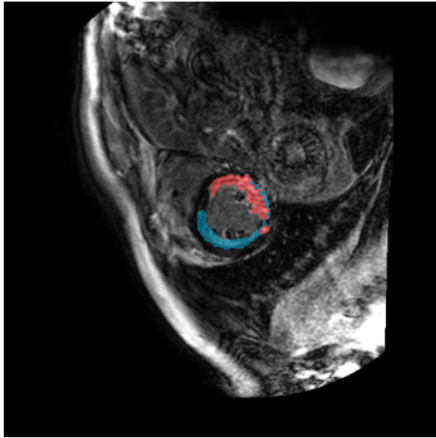
Case3036 | CenterC | C0+LGE+T2 | z=2

nnU-Net

MoSAIC clean

Disagreement

GT

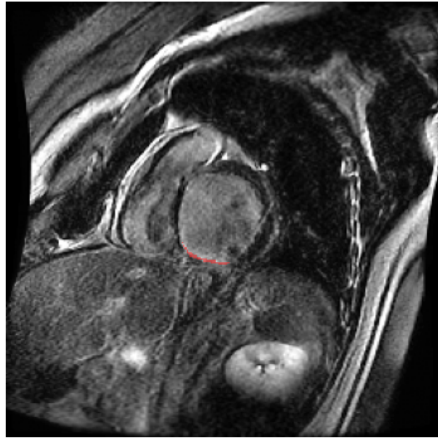


Case1022

center=CenterA modality=LGE-only slice=14

Case1022 | CenterA | LGE-only | z=14

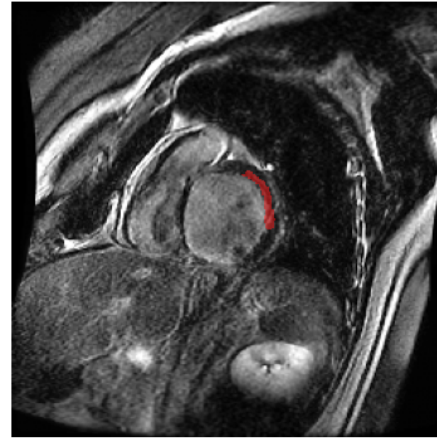
GT



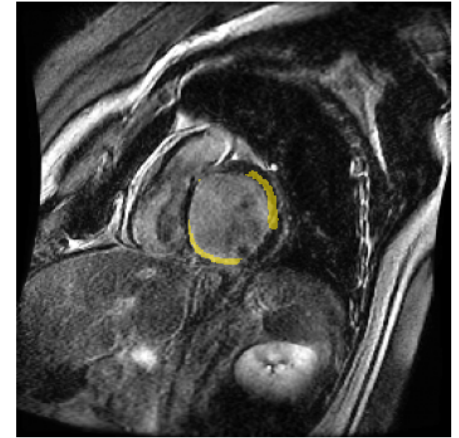
nnU-Net



MoSAIC clean

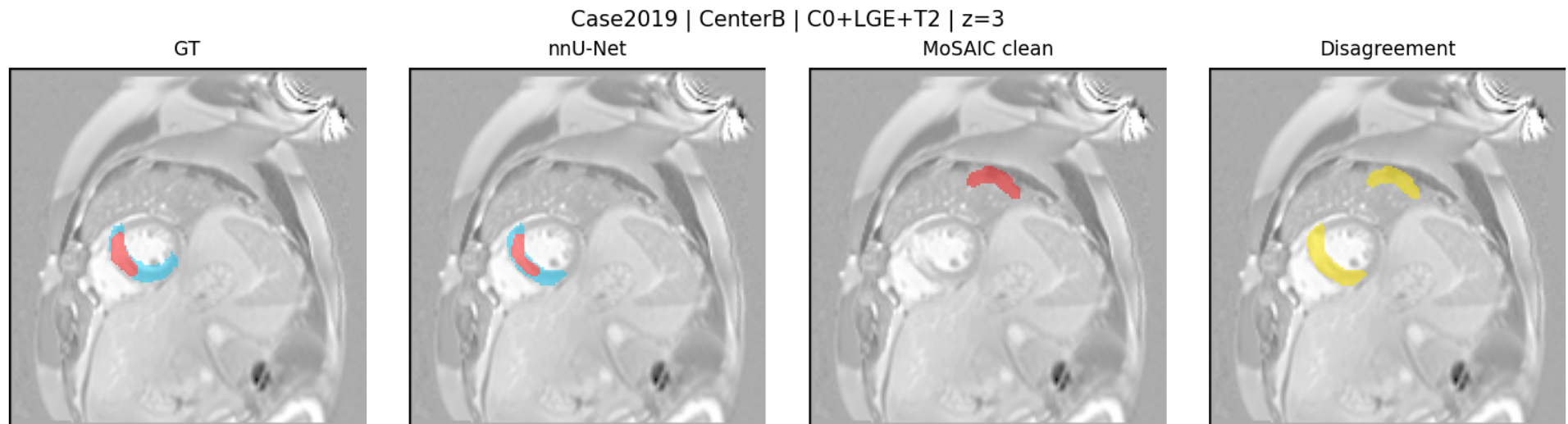


Disagreement



Case2019

center=CenterB modality=C0+LGE+T2 slice=3

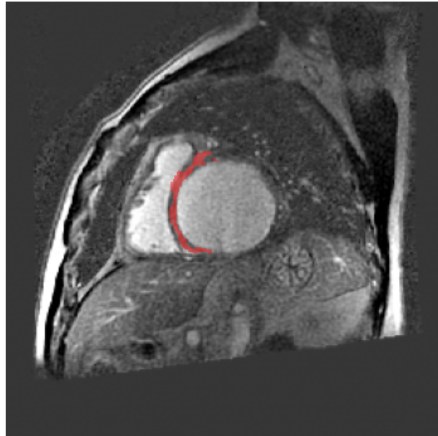


Case1045

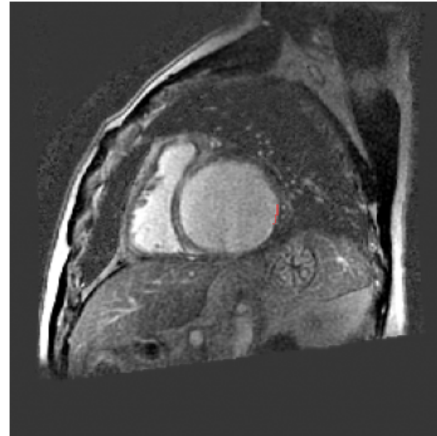
center=CenterA modality=LGE-only slice=6

Case1045 | CenterA | LGE-only | z=6

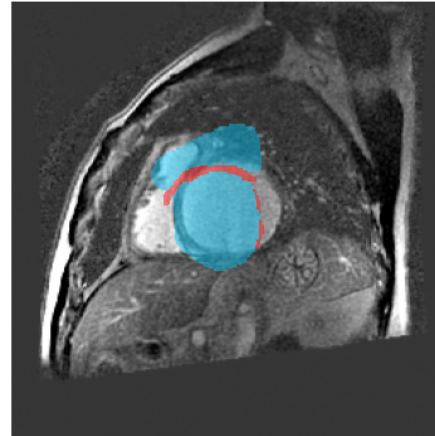
GT



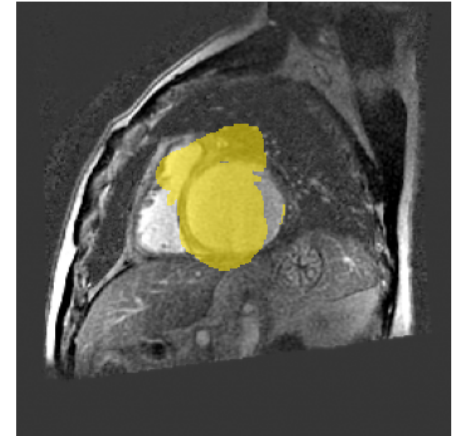
nnU-Net



MoSAIC clean



Disagreement

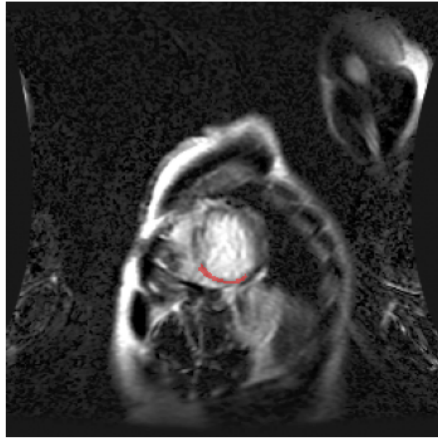


Case1011

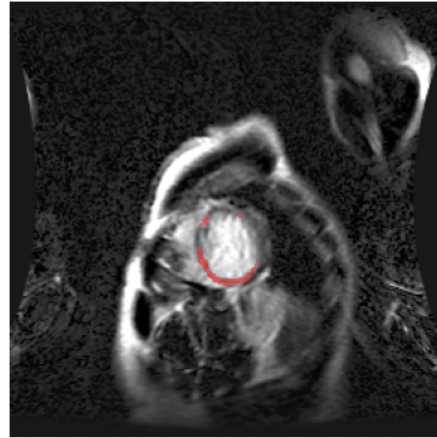
center=CenterA modality=LGE-only slice=1

Case1011 | CenterA | LGE-only | z=1

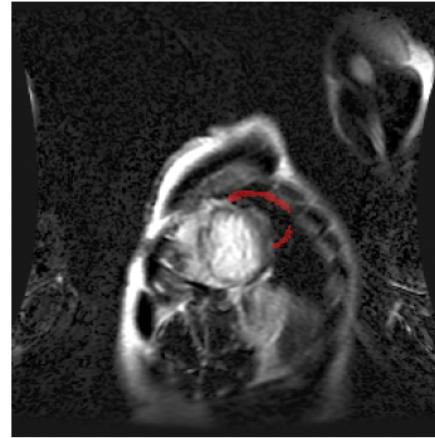
GT



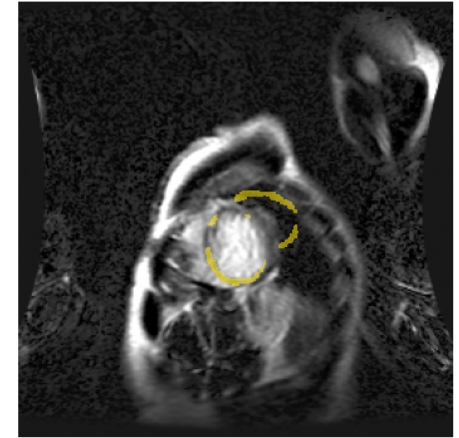
nnU-Net



MoSAIC clean



Disagreement

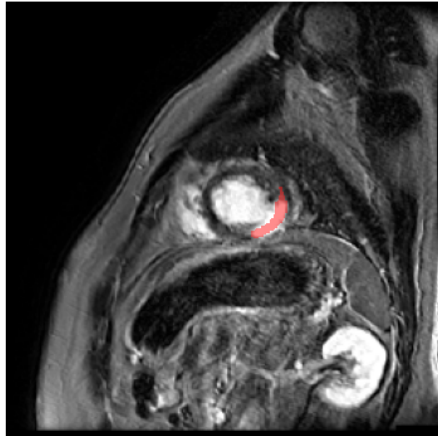


Case1054

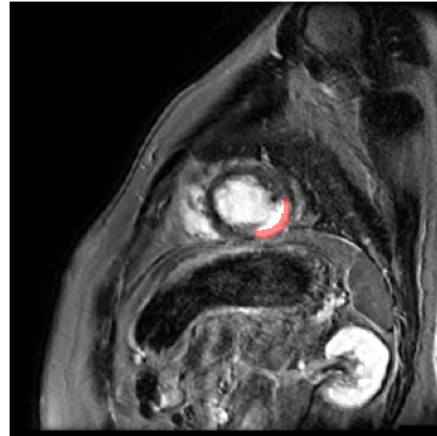
center=CenterA modality=LGE-only slice=7

Case1054 | CenterA | LGE-only | z=7

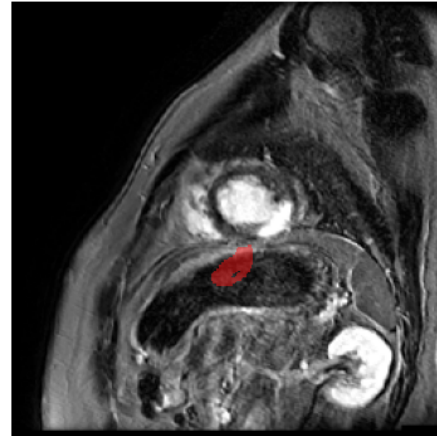
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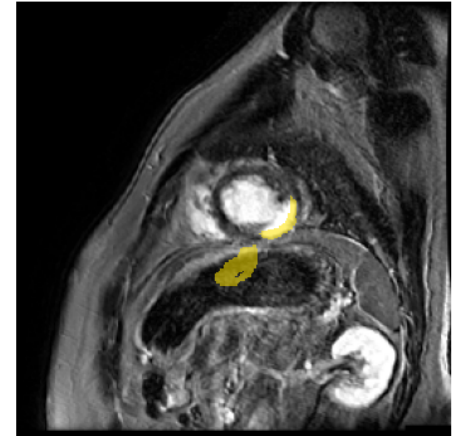
nnU-Net



MoSAIC clean



Disagreement

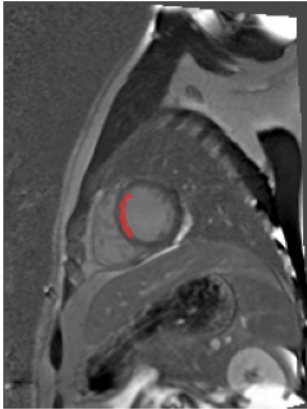


Case8017

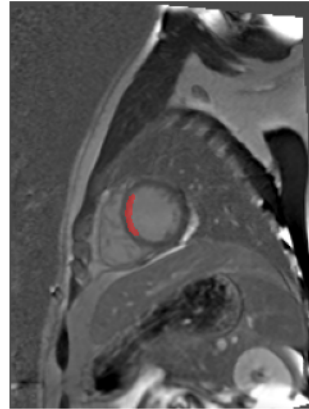
center=CenterH modality=LGE-only slice=5

Case8017 | CenterH | LGE-only | z=5

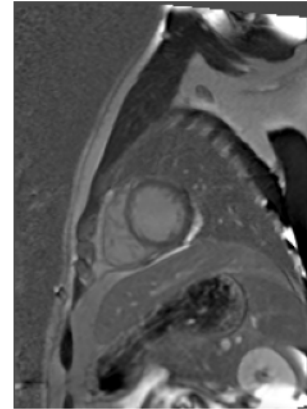
GT



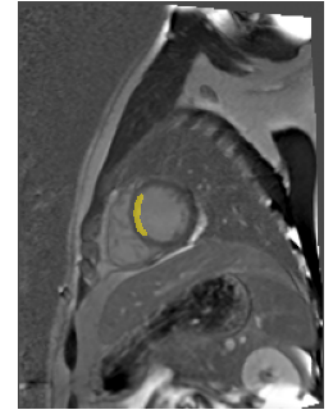
nnU-Net



MoSAIC clean



Disagreement

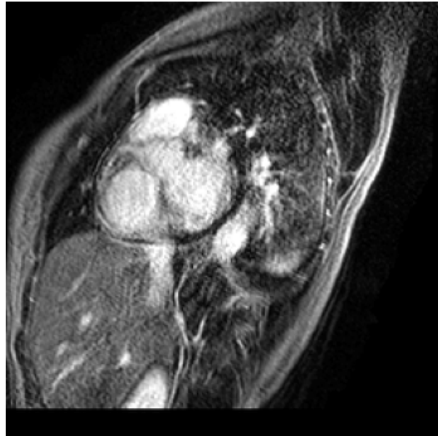


Case1029

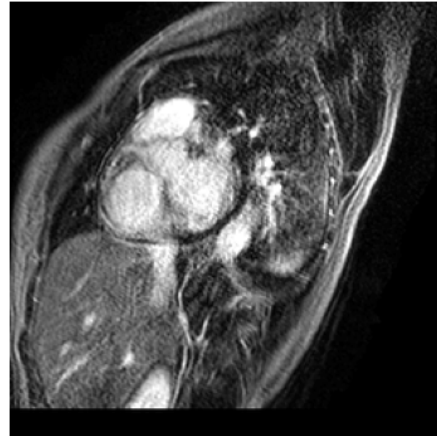
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Case1029 | CenterA | LGE-only | z=17

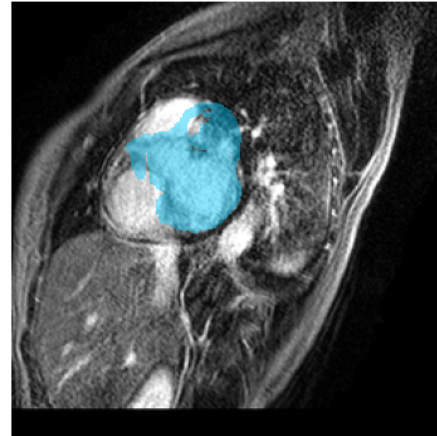
GT



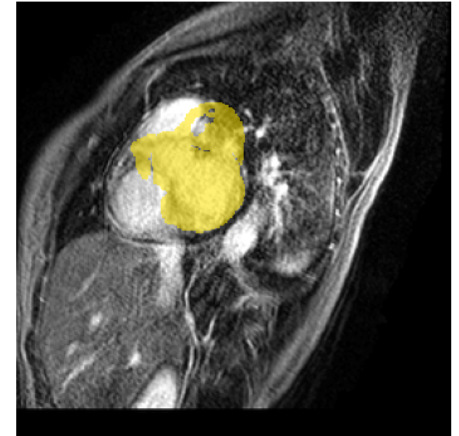
nnU-Net



MoSAIC clean



Disagreement

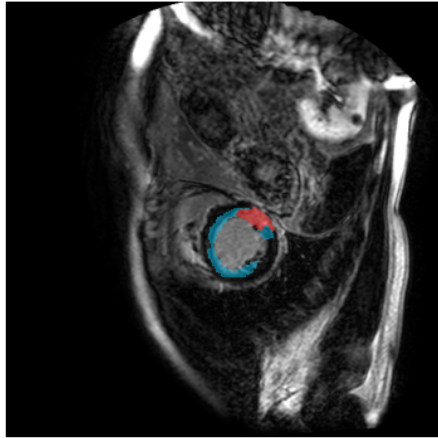


Case3012

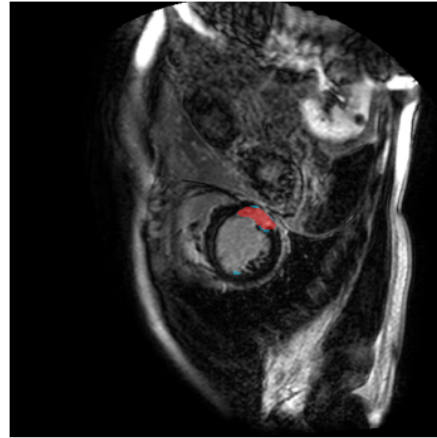
center=CenterC modality=C0+LGE+T2 slice=1

Case3012 | CenterC | C0+LGE+T2 | z=1

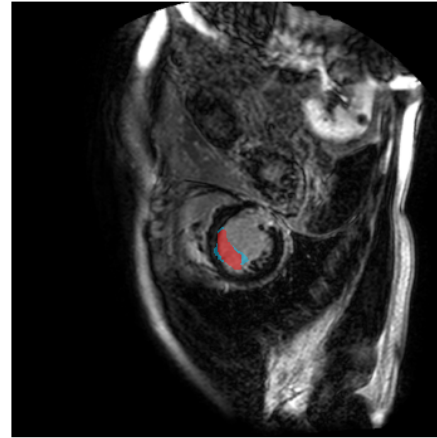
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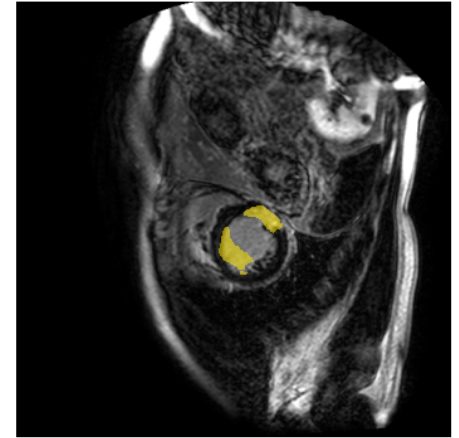
nnU-Net



MoSAIC clean



Disagreement

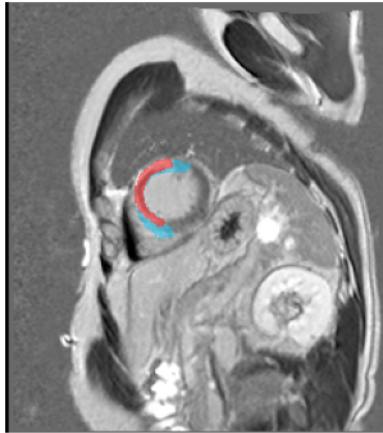


Case2034

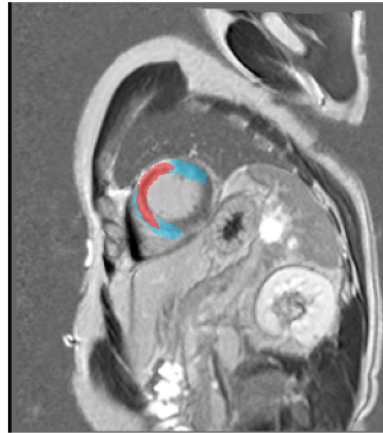
center=CenterB modality=C0+LGE+T2 slice=4

Case2034 | CenterB | C0+LGE+T2 | z=4

GT



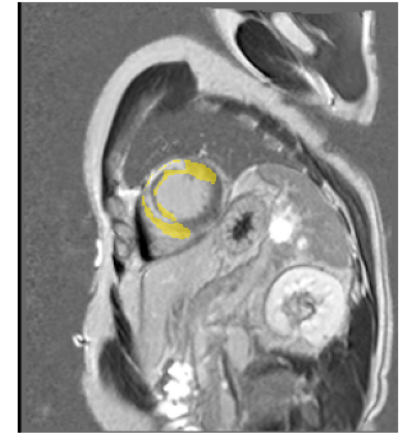
nnU-Net



MoSAIC clean



Disagreement

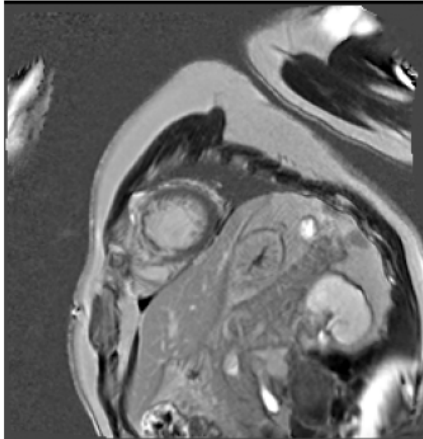


Case2025

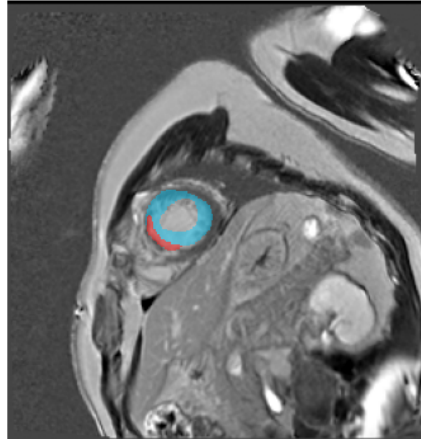
center=CenterB modality=C0+LGE+T2 slice=5

Case2025 | CenterB | C0+LGE+T2 | z=5

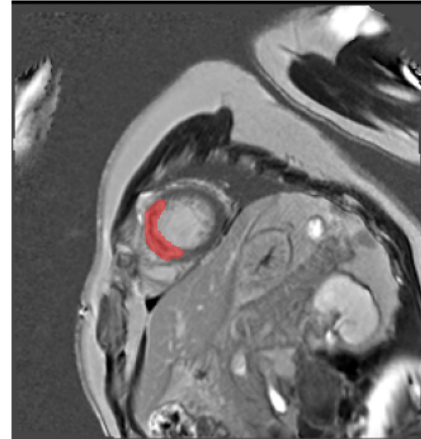
GT



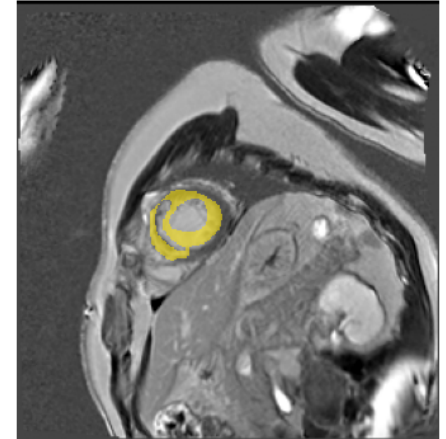
nnU-Net



MoSAIC clean



Disagreement

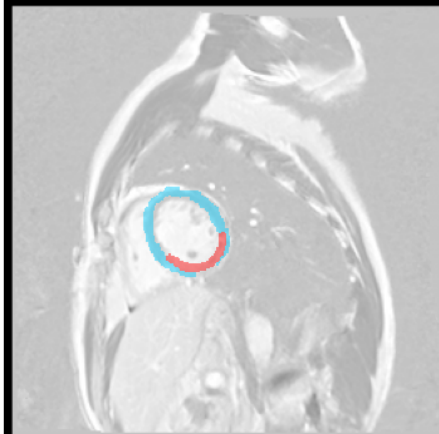


Case2009

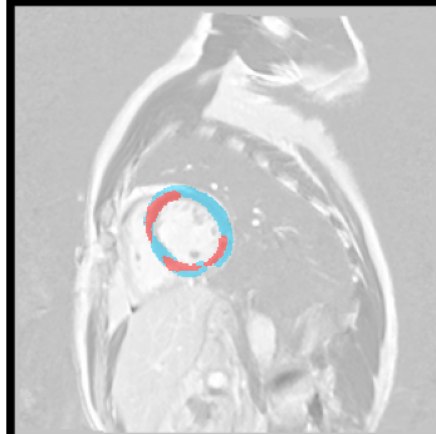
center=CenterB modality=C0+LGE+T2 slice=3

Case2009 | CenterB | C0+LGE+T2 | z=3

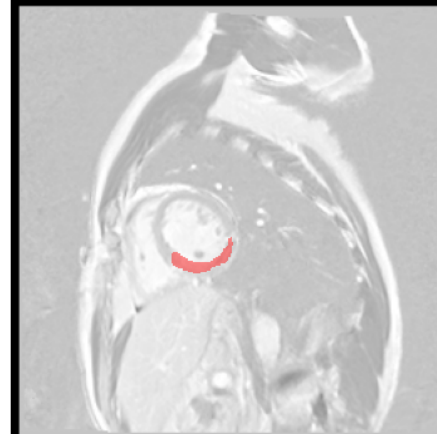
GT



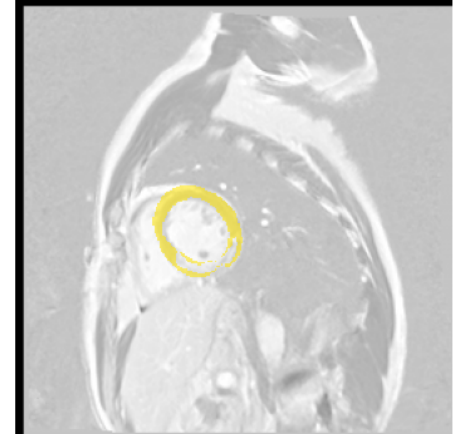
nnU-Net



MoSAIC clean



Disagreement

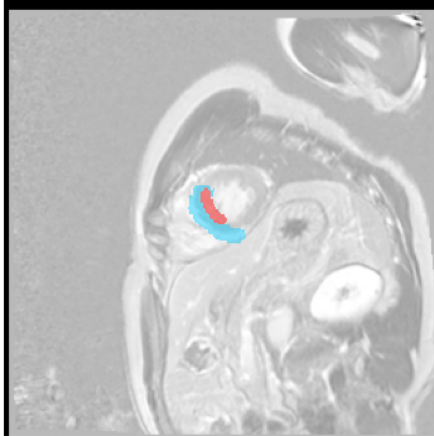


Case2010

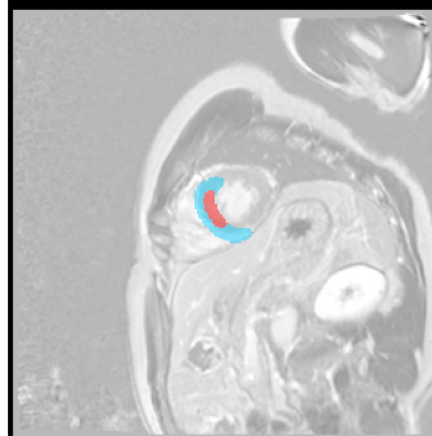
center=CenterB modality=C0+LGE+T2 slice=4

Case2010 | CenterB | C0+LGE+T2 | z=4

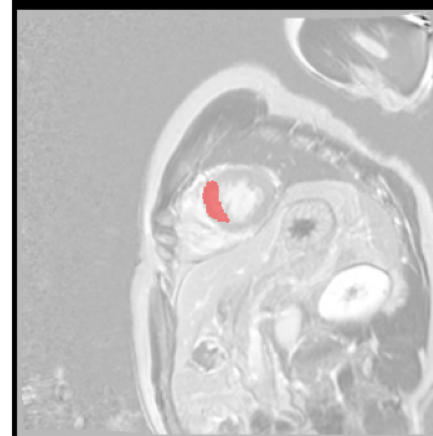
GT



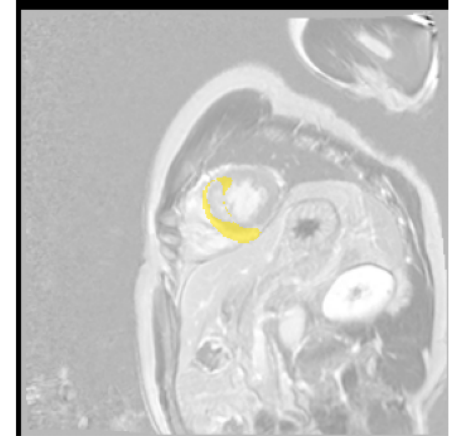
nnU-Net



MoSAIC clean



Disagreement

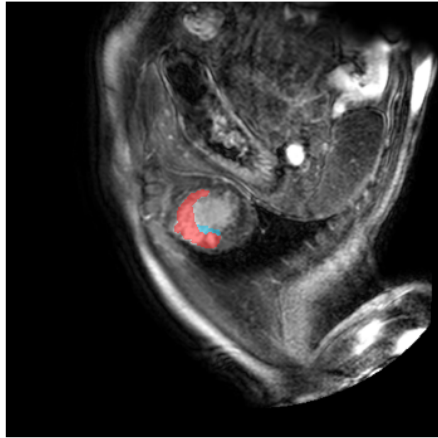


Case3017

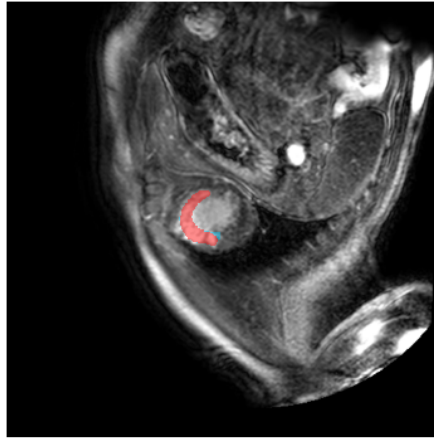
center=CenterC modality=C0+LGE+T2 slice=1

Case3017 | CenterC | C0+LGE+T2 | z=1

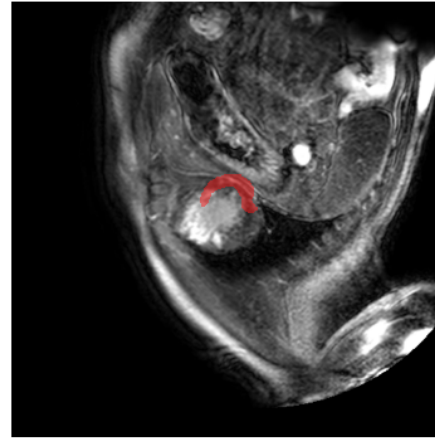
GT



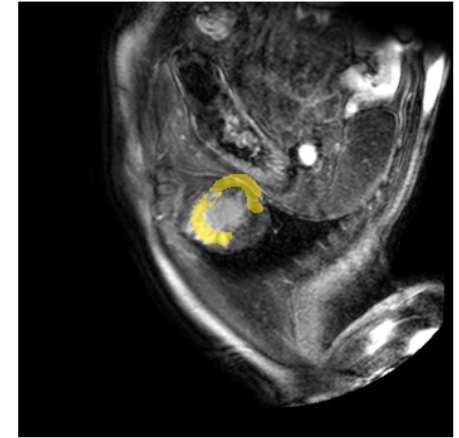
nnU-Net



MoSAIC clean



Disagreement



Case2026

center=CenterB modality=C0+LGE+T2 slice=3

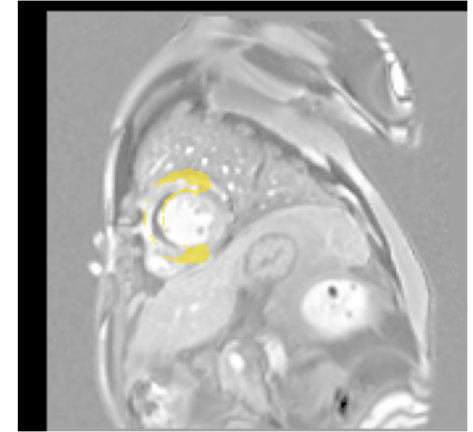
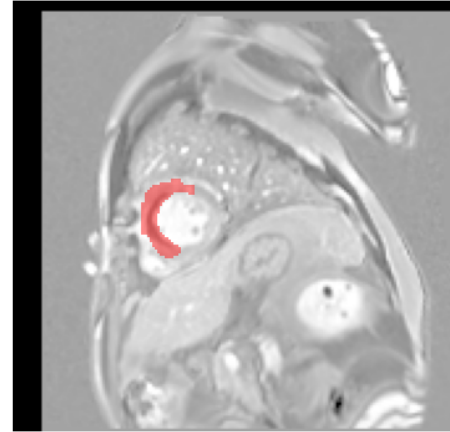
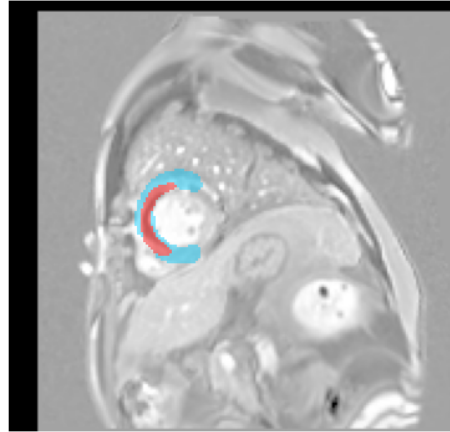
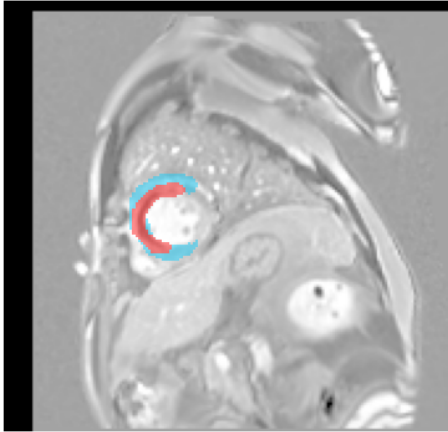
Case2026 | CenterB | C0+LGE+T2 | z=3

GT

nnU-Net

MoSAIC clean

Disagreement



Case1033

center=CenterA modality=LGE-only slice=16

Case1033 | CenterA | LGE-only | z=16

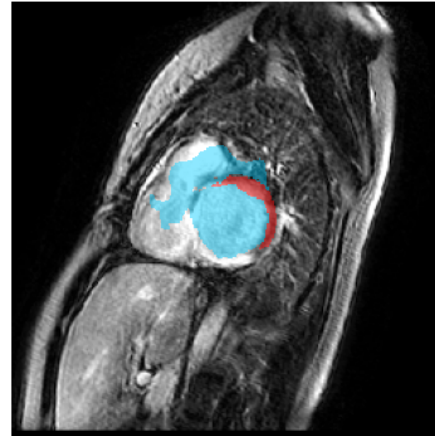
GT



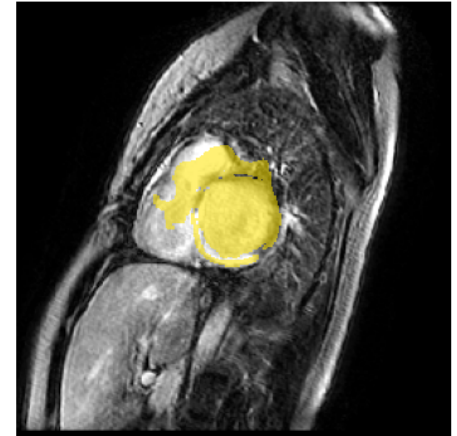
nnU-Net



MoSAIC clean



Disagreement

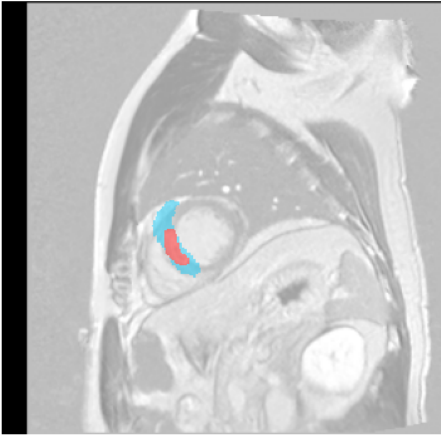


Case2029

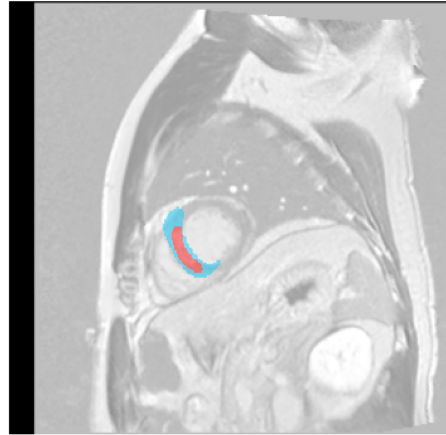
center=CenterB modality=C0+LGE+T2 slice=5

Case2029 | CenterB | C0+LGE+T2 | z=5
nnU-Net MoSAIC clean

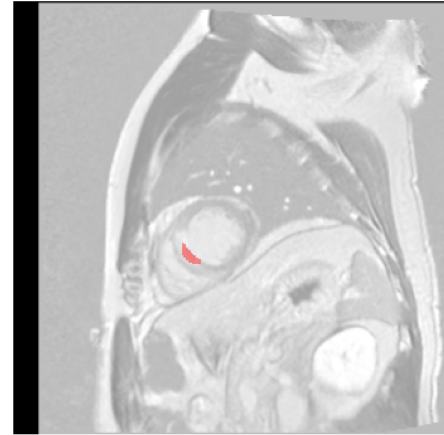
GT



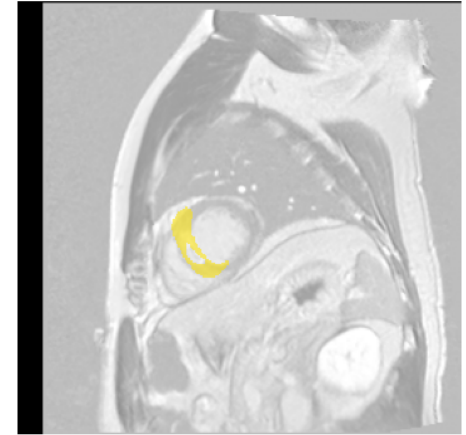
nnU-Net



MoSAIC clean



Disagreement

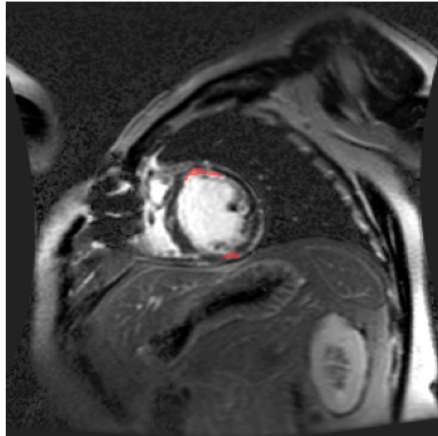


Case1014

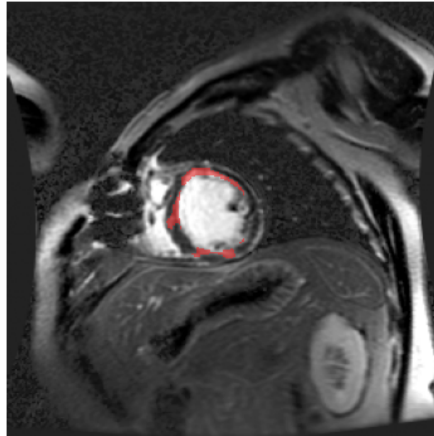
center=CenterA modality=LGE-only slice=4

Case1014 | CenterA | LGE-only | z=4

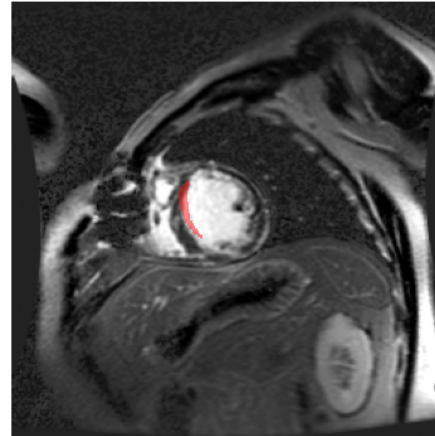
GT



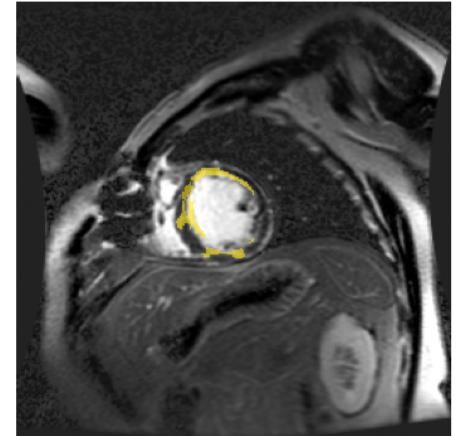
nnU-Net



MoSAIC clean



Disagreement



17. 错误重合和模型互补上限

case oracle 对 nnU-Net 的直接提升很小，scar 约 0.022、pure edema 约 0.002、lesion union 约 0.013；voxel TP oracle 很高，但这是不可部署上限，不能当作模型性能。selector feasibility 显示 scar 有病例级可辨识信号，pure edema 证据弱。

metric_name	case_gain	voxel_gain	selector_signal
scar	0.022	0.2375	0.8269
pure_edema	0.0023	0.173	0
lesion_union	0.0133	0.2247	0

18. selector feasibility

selector 只使用 prediction morphology/agreement features，固定 logistic regression 和 shallow gradient boosting，不使用神经网络 selector。scar selector AUROC 约 0.827；pure edema 因 MoSAIC-better 正例过少而阻塞。

19. 冻结特征可分性 probe

第 19 页使用窄表/短字段，不使用会溢出的宽表。V2 绑定 MoSAIC coarse/scar fine component features 与 raw intensity controls; nnU-Net/PRISM frozen activation 未导出，因此按缺资产阻塞，不伪造成无信号。

feature_source	status	n_artifacts	limit
nnU-Net encoder	MISSING_ASSET	0	No bound frozen activ...
nnU-Net decoder	MISSING_ASSET	0	No bound frozen activ...
PRISM shared	MISSING_ASSET	0	No bound frozen activ...
PRISM routed	MISSING_ASSET	0	No bound frozen activ...
PRISM refiner	MISSING_ASSET	0	No bound frozen activ...
MoSAIC coarse	VALID	5	Does not cover pure-e...
MoSAIC scar fine	VALID	5	Does not cover pure-e...
MoSAIC edema	MISSING_ASSET	0	No bound frozen activ...
raw intensity control	VALID	220	Control is case-level...

20. decoder-reset 诊断对照

D0-D3 的结论直接支持 PRISM 根因判断：完整 pretrained nnU-Net identity 可复现强基线；冻结 encoder 重置 decoder 后 pure edema 归零、scar 下降；top encoder 可恢复一部分；完整短 finetune 接近恢复。这说明 decoder/训练 recipe 是核心，不是只要 encoder 迁移就够。

21-24. alignment、scar、pure edema 和 Cine

alignment 绑定 20260703 complete-case 诊断，未支持多序列错位是主因。Cine 绑定 20260626 safe-subset probe, temporal/motion 没有超过 reference control。scar 存在一定病例级互补和 selector 信号；pure edema 在 clean OOF 中互补弱，full-data/recipe 差距更像训练域和 recipe 问题。

25. 为什么过去多次充分设计仍然失败

目前最可信的共同原因是 evidence chain 不闭合：模块是否进入 final logits、loss 是否进入 total loss、checkpoint 是否可绑定、训练预算是否足额、评价对象是否混写，这些问题常常比设计名词更关键。组件生存清单把“思想有效、实现失败、未验证、思想失败”分开记录。

source_model	component	future	repeat_risk
Batch7	availability-aware ev...	KEEP_DATA_RULE	medium
Batch7	pathology-specific re...	RETEST_OPTION	high
Batch7	negative-space	RETEST_OPTION	high
Batch7	complex router/SIP	DO_NOT_REUSE_CURRENT_...	high
MMRD	reliable-label superv...	KEEP_DATA_RULE	low
MMRD	no-T2 edema hygiene	KEEP_DATA_RULE	low
MMRD	simple residual patho...	DO_NOT_REUSE_CURRENT_...	high
Cascade	strong baseline fallback	KEEP_STRONG	low
Cascade	bounded correction	RETEST_OPTION	medium
Cascade	prototype input	UNRESOLVED	high
ARC	direct reconstruction	RETEST_OPTION	medium
ARC	decoder reset	DO_NOT_REUSE	high
DG/DR/DPR	pathology-specific ar...	RETEST_OPTION	medium
PRISM	private pyramids/routing	DO_NOT_REUSE_CURRENT_...	high
PRISM	stage schedule	RETEST_OPTION	medium

26. 根因排序与证据图

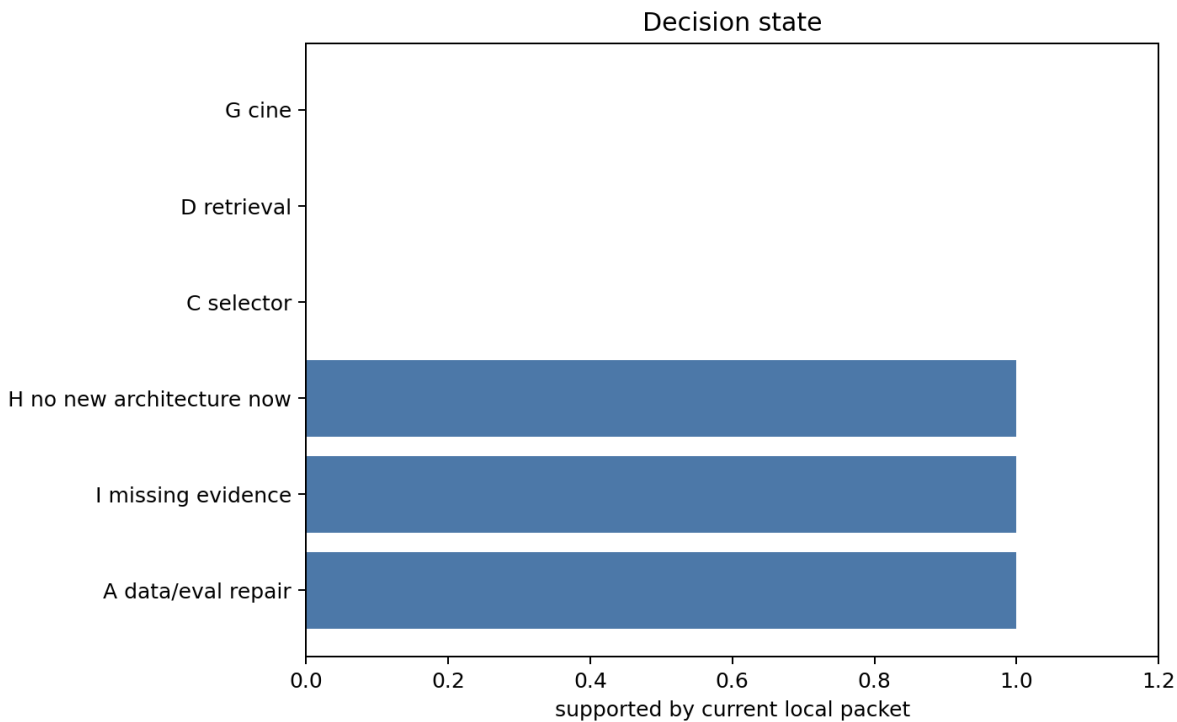


Figure 5: 决策状态

- **METRIC_IMPLEMENTATION**
 - severity: HIGH
 - confidence: MODERATE
 - confirmed: True

- evidence: remote FP 和 pure-edema/edema-zone 语义已有 known-bad 保护；全量影响未重算。
- **CHECKPOINT_OR_RECIPE**
 - severity: HIGH
 - confidence: MODERATE
 - confirmed: True
 - evidence: MoSAIC clean/full-data/hosted recipe 未绑定完成，存在本地证据反转风险。
- **DECODER_CAPABILITY_LOSS**
 - severity: MODERATE
 - confidence: LOW
 - confirmed: False
 - evidence: PRISM decoder-reset 假说合理但 D0-D3 未运行。
- **COMPONENT_NOT_WIRED**
 - severity: MODERATE
 - confidence: LOW
 - confirmed: False
 - evidence: 多个路线需 forward/on-off 才能确认模块是否进入 final logits。
- **INSUFFICIENT_PATHOLOGY_SIGNAL**
 - severity: MODERATE
 - confidence: UNRESOLVED
 - confirmed: False
 - evidence: feature probe 未运行。
- **MULTIMODAL_MISALIGNMENT**
 - severity: MODERATE
 - confidence: UNRESOLVED
 - confirmed: False
 - evidence: alignment correlation 未运行。
- **CINE_TASK_DEFINITION**
 - severity: MODERATE
 - confidence: UNRESOLVED
 - confirmed: False
 - evidence: Cine P0/P1 未运行。

27. 当前能下的结论

Local Evidence Conclusions

当前本地证据支持 A 和 I: 先做评价/数据/recipe 绑定修复, 并承认关键证据仍缺失。尚不能支持新的 CARE 架构蓝图。

28. 当前不能下的结论

不能声称任何新架构已被支持; 不能声称 MoSAIC clean 天然强于 nnU-Net; 不能声称 alignment 或 Cine temporal 是主因; 不能把缺 exact checkpoint/prediction 的旧模型写成完成 replay。

29. 外部 Deep Research 必须回答的问题

- [DR-001] Small-lesion scar segmentation beyond nnU-Net requires which evidence standard?
- [DR-002] Can clean MoSAIC recipe gains be separated from full-data target-domain advantage?
- [DR-003] Do frozen encoder features contain patient-held-out scar FN/FP separability?
- [DR-004] When does cine temporal information improve pathology segmentation over ED-only?

30. 下一轮决策树

Research Decision Tree

1. 先完成 evaluation/data repair。
2. 若 D0 不能复现 nnU-Net, 停止 decoder-reset。
3. 若 selector nested CV 不超过 always-best-single-model, 停止 deployable selector。
4. 若 feature probe control 不成立, 停止 retrieval/prototype 叙事。

附录 A: 模型和 checkpoint provenance

checkpoint 清单只显示定位字段, 不展开长路径列, 避免右侧截断。完整路径仍保留在 CSV。

model_id	size_bytes	hash_status	quality
NNUNET	354608437	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	354266799	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	354383029	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	354382767	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	354382965	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	354201839	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	354383093	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	354242031	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	354383349	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	354270127	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	354382965	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	354369135	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	31579	PREFIX_8192_BYTES	PATH_BOUND
NNUNET	235538772	LARGE_METADATA_ONLY	PATH_BOUND
NNUNET	4305	FULL	PATH_BOUND

checkpoint binding 以聚合计数进入 PDF;完整逐项路径保留在 historical_checkpoint_binding.csv。

model_id	artifact	status	rows
B0_3_SRR_ANCHOR	checkpoint	BOUND	3
B7_BR2_SIP	checkpoint	BOUND	21
CARE_ARC	checkpoint	BOUND	8
DG_DR_DPR	checkpoint	BOUND	120
DG_DR_DPR	checkpoint	BOUND_TRUNCATED_FOR_P...	1
MMRD_BATCH9	checkpoint	BOUND	9
PRISM_V2	checkpoint	BOUND	13
SRR_CASCADE	checkpoint	BOUND	17

附录 B: 指标公式和 known-bad

- **E-DATA-001**
 - source_path: data_case_manifest.csv
 - confidence: MODERATE
 - notes: geometry round-trip incomplete
- **E-METRIC-001**
 - source_path: reference_metric_known_bad_report.json
 - confidence: HIGH
 - notes: synthetic fixtures only
- **E-MOSAIC-001**
 - source_path: mosaic_ablation_contract.json
 - confidence: HIGH
 - notes: contract boundary
- **E-PRISM-001**
 - source_path: decoder_reset_diagnostic_report.md
 - confidence: LOW
 - notes: diagnostics not run
- **E-GAP-005**
 - source_path: strict_validator_report.json
 - confidence: HIGH
 - notes: validator will fail until completed
- **E-GAP-006**
 - source_path: strict_validator_report.json
 - confidence: HIGH
 - notes: validator will fail until completed
- **E-GAP-007**
 - source_path: strict_validator_report.json
 - confidence: HIGH
 - notes: validator will fail until completed
- **E-GAP-008**
 - source_path: strict_validator_report.json
 - confidence: HIGH
 - notes: validator will fail until completed
- **E-GAP-009**
 - source_path: strict_validator_report.json
 - confidence: HIGH
 - notes: validator will fail until completed
- **E-GAP-010**
 - source_path: strict_validator_report.json
 - confidence: HIGH
 - notes: validator will fail until completed
- **E-GAP-011**
 - source_path: strict_validator_report.json
 - confidence: HIGH
 - notes: validator will fail until completed
- **E-GAP-012**
 - source_path: strict_validator_report.json
 - confidence: HIGH
 - notes: validator will fail until completed

附录 C: Slurm 和运行回执

本次 PDF 重渲染没有提交新的 Slurm job。已有 packet 的 controller context 和 V2 GPU manifest 记录了启动时可见的 Slurm 状态、G1-G4 GPU steps 与 G5-G10 聚合状态。

time_utc	phase	decision	next_action
07-30 02:42:43	F0	PARTIAL_BOOTSTRAP	build inventories and...
07-30 02:49:59	F0	PARTIAL_BOOTSTRAP	build inventories and...
07-30 02:51:37	F0	PARTIAL_BOOTSTRAP	build inventories and...
07-30 02:53:12	F0	PARTIAL_BOOTSTRAP	build inventories and...
07-30 02:54:04	F0	PARTIAL_BOOTSTRAP	build inventories and...
07-30 02:55:51	F0	PARTIAL_BOOTSTRAP	build inventories and...
07-30 02:57:24	F0	PARTIAL_BOOTSTRAP	build inventories and...
07-30 04:03:17	F0B_READY	NEEDS_D0	run D0 identity repla...
07-30 04:10:19	F7B_D0	D0_PASS_D1_D3_READY	run decoder-reset dia...

附录 D：完整病例级表格索引

完整病例级重聚合已在 V2 可绑定证据范围内完成。这里列出机器可读表的位置和状态，不展开长路径列，避免 PDF 裁切。

file	status
standardized_casewise_metrics.csv	DONE_OOF
case_oracle_summary.csv	DONE_OOF
historical_result_comparability.csv	DONE_HIST_METRIC
prism_corrected_casewise_metrics.csv	DONE_PRISM13

附录 E1: standardized casewise metrics 分块

case_id	center	metric_name	model_id	dice	empty_pred
Case1002	CenterA	scar	nnunet_oof	0.6167	0
Case1002	CenterA	scar	mosaic_oof	0.1155	0
Case1002	CenterA	lesion_union	nnunet_oof	0.6167	0
Case1002	CenterA	lesion_union	mosaic_oof	0.1783	0
Case1007	CenterA	scar	nnunet_oof	0.5811	0
Case1007	CenterA	scar	mosaic_oof	0.2493	0
Case1007	CenterA	lesion_union	nnunet_oof	0.5811	0
Case1007	CenterA	lesion_union	mosaic_oof	0.0841	0

附录 E1: standardized casewise metrics 分块 (续 2)

case_id	center	metric_name	model_id	dice	empty_pred
Case1009	CenterA	scar	nnunet_oof	0.6049	0
Case1009	CenterA	scar	mosaic_oof	0.0863	0
Case1009	CenterA	lesion_union	nnunet_oof	0.6049	0
Case1009	CenterA	lesion_union	mosaic_oof	0.1403	0
Case1010	CenterA	scar	nnunet_oof	0.4828	0
Case1010	CenterA	scar	mosaic_oof	0.252	0
Case1010	CenterA	lesion_union	nnunet_oof	0.4828	0
Case1010	CenterA	lesion_union	mosaic_oof	0.1303	0

附录 E1: standardized casewise metrics 分块 (续 3)

case_id	center	metric_name	model_id	dice	empty_pred
Case1021	CenterA	scar	nnunet_oof	0.5699	0
Case1021	CenterA	scar	mosaic_oof	0.3916	0
Case1021	CenterA	lesion_union	nnunet_oof	0.5699	0
Case1021	CenterA	lesion_union	mosaic_oof	0.2041	0
Case1023	CenterA	scar	nnunet_oof	0.6466	0
Case1023	CenterA	scar	mosaic_oof	0.0666	0
Case1023	CenterA	lesion_union	nnunet_oof	0.6466	0
Case1023	CenterA	lesion_union	mosaic_oof	0.0624	0

附录 E1: standardized casewise metrics 分块 (续 4)

case_id	center	metric_name	model_id	dice	empty_pred
Case1029	CenterA	scar	nnunet_oof	0.1644	0
Case1029	CenterA	scar	mosaic_oof	0.0012	0
Case1029	CenterA	lesion_union	nnunet_oof	0.1644	0
Case1029	CenterA	lesion_union	mosaic_oof	0.002	0
Case1033	CenterA	scar	nnunet_oof	0.6714	0
Case1033	CenterA	scar	mosaic_oof	0.0963	0
Case1033	CenterA	lesion_union	nnunet_oof	0.6714	0
Case1033	CenterA	lesion_union	mosaic_oof	0.1057	0

附录 E1: standardized casewise metrics 分块 (续 5)

case_id	center	metric_name	model_id	dice	empty_pred
Case1040	CenterA	scar	nnunet_oof	0.428	0
Case1040	CenterA	scar	mosaic_oof	0.1113	0
Case1040	CenterA	lesion_union	nnunet_oof	0.428	0
Case1040	CenterA	lesion_union	mosaic_oof	0.0843	0
Case1042	CenterA	scar	nnunet_oof	0.7574	0
Case1042	CenterA	scar	mosaic_oof	0.0667	0
Case1042	CenterA	lesion_union	nnunet_oof	0.7574	0
Case1042	CenterA	lesion_union	mosaic_oof	0.0416	0

附录 E1: standardized casewise metrics 分块 (续 6)

case_id	center	metric_name	model_id	dice	empty_pred
Case1045	CenterA	scar	nnunet_oof	0.0997	0
Case1045	CenterA	scar	mosaic_oof	0.0657	0
Case1045	CenterA	lesion_union	nnunet_oof	0.0997	0
Case1045	CenterA	lesion_union	mosaic_oof	0.0997	0
Case1047	CenterA	scar	nnunet_oof	0.6283	0
Case1047	CenterA	scar	mosaic_oof	0.4432	0
Case1047	CenterA	lesion_union	nnunet_oof	0.6283	0
Case1047	CenterA	lesion_union	mosaic_oof	0.2364	0

附录 E1: standardized casewise metrics 分块 (续 7)

case_id	center	metric_name	model_id	dice	empty_pred
Case1053	CenterA	scar	nnunet_oof	0.5196	0
Case1053	CenterA	scar	mosaic_oof	0.1174	0
Case1053	CenterA	lesion_union	nnunet_oof	0.5196	0
Case1053	CenterA	lesion_union	mosaic_oof	0.0532	0
Case1062	CenterA	scar	nnunet_oof	0.5071	0
Case1062	CenterA	scar	mosaic_oof	0.1835	0
Case1062	CenterA	lesion_union	nnunet_oof	0.5071	0
Case1062	CenterA	lesion_union	mosaic_oof	0.0974	0

附录 E1: standardized casewise metrics 分块 (续 8)

case_id	center	metric_name	model_id	dice	empty_pred
Case1070	CenterA	scar	nnunet_oof	0.5005	0
Case1070	CenterA	scar	mosaic_oof	0.4024	0
Case1070	CenterA	lesion_union	nnunet_oof	0.5005	0
Case1070	CenterA	lesion_union	mosaic_oof	0.0935	0
Case1073	CenterA	scar	nnunet_oof	0.4283	0
Case1073	CenterA	scar	mosaic_oof	0.1166	0
Case1073	CenterA	lesion_union	nnunet_oof	0.4283	0
Case1073	CenterA	lesion_union	mosaic_oof	0.1266	0

附录 E1: standardized casewise metrics 分块 (续 9)

case_id	center	metric_name	model_id	dice	empty_pred
Case1080	CenterA	scar	nnunet_oof	0.6369	0
Case1080	CenterA	scar	mosaic_oof	0.1006	0
Case1080	CenterA	lesion_union	nnunet_oof	0.6369	0
Case1080	CenterA	lesion_union	mosaic_oof	0.0738	0
Case2002	CenterB	scar	nnunet_oof	0.5603	0
Case2002	CenterB	scar	mosaic_oof	0.7486	0
Case2002	CenterB	pure_edema	nnunet_oof	0.5376	0
Case2002	CenterB	pure_edema	mosaic_oof	0.446	0

附录 E1: standardized casewise metrics 分块 (续 10)

case_id	center	metric_name	model_id	dice	empty_pred
Case2002	CenterB	lesion_union	nnunet_oof	0.6753	0
Case2002	CenterB	lesion_union	mosaic_oof	0.5789	0
Case2007	CenterB	scar	nnunet_oof	0.4899	0
Case2007	CenterB	scar	mosaic_oof	0.5998	0
Case2007	CenterB	pure_edema	nnunet_oof	0.5701	0
Case2007	CenterB	pure_edema	mosaic_oof	0.5136	0
Case2007	CenterB	lesion_union	nnunet_oof	0.7221	0
Case2007	CenterB	lesion_union	mosaic_oof	0.7312	0

附录 E1: standardized casewise metrics 分块 (续 11)

case_id	center	metric_name	model_id	dice	empty_pred
Case2008	CenterB	scar	nnunet_oof	0.7532	0
Case2008	CenterB	scar	mosaic_oof	0.7747	0
Case2008	CenterB	pure_edema	nnunet_oof	0.3486	0
Case2008	CenterB	pure_edema	mosaic_oof	0.2999	0
Case2008	CenterB	lesion_union	nnunet_oof	0.5703	0
Case2008	CenterB	lesion_union	mosaic_oof	0.5597	0
Case2017	CenterB	scar	nnunet_oof	0.5471	0
Case2017	CenterB	scar	mosaic_oof	0.6924	0

附录 E1: standardized casewise metrics 分块 (续 12)

case_id	center	metric_name	model_id	dice	empty_pred
Case2017	CenterB	pure_edema	nnunet_oof	0.6796	0
Case2017	CenterB	pure_edema	mosaic_oof	0.381	0
Case2017	CenterB	lesion_union	nnunet_oof	0.8161	0
Case2017	CenterB	lesion_union	mosaic_oof	0.5424	0
Case2020	CenterB	scar	nnunet_oof	0.4196	0
Case2020	CenterB	scar	mosaic_oof	0	0
Case2020	CenterB	pure_edema	nnunet_oof	0.5631	0
Case2020	CenterB	pure_edema	mosaic_oof	0	0

附录 E2: case oracle 和 voxel oracle 分块

case_id	center	metric_name	best_model	case_oracle	voxel_oracle
Case1002	CenterA	scar	nnunet_oof	0.6167	0.7338
Case1002	CenterA	lesion_union	nnunet_oof	0.6167	0.8947
Case1007	CenterA	scar	nnunet_oof	0.5811	0.912
Case1007	CenterA	lesion_union	nnunet_oof	0.5811	0.9487
Case1009	CenterA	scar	nnunet_oof	0.6049	0.7778
Case1009	CenterA	lesion_union	nnunet_oof	0.6049	0.8556
Case1010	CenterA	scar	nnunet_oof	0.4828	0.789
Case1010	CenterA	lesion_union	nnunet_oof	0.4828	0.9466

附录 E2: case oracle 和 voxel oracle 分块 (续 2)

case_id	center	metric_name	best_model	case_oracle	voxel_oracle
Case1021	CenterA	scar	nnunet_oof	0.5699	0.8076
Case1021	CenterA	lesion_union	nnunet_oof	0.5699	0.8911
Case1023	CenterA	scar	nnunet_oof	0.6466	0.8091
Case1023	CenterA	lesion_union	nnunet_oof	0.6466	0.9032
Case1029	CenterA	scar	nnunet_oof	0.1644	0.8728
Case1029	CenterA	lesion_union	nnunet_oof	0.1644	0.9024
Case1033	CenterA	scar	nnunet_oof	0.6714	0.9046
Case1033	CenterA	lesion_union	nnunet_oof	0.6714	0.9621

附录 E2: case oracle 和 voxel oracle 分块 (续 3)

case_id	center	metric_name	best_model	case_oracle	voxel_oracle
Case1040	CenterA	scar	nnunet_oof	0.428	0.7495
Case1040	CenterA	lesion_union	nnunet_oof	0.428	0.9095
Case1042	CenterA	scar	nnunet_oof	0.7574	0.8738
Case1042	CenterA	lesion_union	nnunet_oof	0.7574	0.8985
Case1045	CenterA	scar	nnunet_oof	0.0997	0.219
Case1045	CenterA	lesion_union	mosaic_oof	0.0997	0.9229
Case1047	CenterA	scar	nnunet_oof	0.6283	0.8
Case1047	CenterA	lesion_union	nnunet_oof	0.6283	0.9153

附录 E2: case oracle 和 voxel oracle 分块 (续 4)

case_id	center	metric_name	best_model	case_oracle	voxel_oracle
Case1053	CenterA	scar	nnunet_oof	0.5196	0.6928
Case1053	CenterA	lesion_union	nnunet_oof	0.5196	0.7743
Case1062	CenterA	scar	nnunet_oof	0.5071	0.8914
Case1062	CenterA	lesion_union	nnunet_oof	0.5071	0.9571
Case1070	CenterA	scar	nnunet_oof	0.5005	0.9154
Case1070	CenterA	lesion_union	nnunet_oof	0.5005	0.9742
Case1073	CenterA	scar	nnunet_oof	0.4283	0.6056
Case1073	CenterA	lesion_union	nnunet_oof	0.4283	0.811

附录 E2: case oracle 和 voxel oracle 分块 (续 5)

case_id	center	metric_name	best_model	case_oracle	voxel_oracle
Case1080	CenterA	scar	nnunet_oof	0.6369	0.8048
Case1080	CenterA	lesion_union	nnunet_oof	0.6369	0.8946
Case2002	CenterB	scar	mosaic_oof	0.7486	0.9009
Case2002	CenterB	pure_edema	nnunet_oof	0.5376	0.8198
Case2002	CenterB	lesion_union	nnunet_oof	0.6753	0.8741
Case2007	CenterB	scar	mosaic_oof	0.5998	0.7985
Case2007	CenterB	pure_edema	nnunet_oof	0.5701	0.9036
Case2007	CenterB	lesion_union	mosaic_oof	0.7312	0.9473

附录 E2: case oracle 和 voxel oracle 分块 (续 6)

case_id	center	metric_name	best_model	case_oracle	voxel_oracle
Case2008	CenterB	scar	mosaic_oof	0.7747	0.9579
Case2008	CenterB	pure_edema	nnunet_oof	0.3486	0.4639
Case2008	CenterB	lesion_union	nnunet_oof	0.5703	0.6719
Case2017	CenterB	scar	mosaic_oof	0.6924	0.8995
Case2017	CenterB	pure_edema	nnunet_oof	0.6796	0.9098
Case2017	CenterB	lesion_union	nnunet_oof	0.8161	0.9754
Case2020	CenterB	scar	nnunet_oof	0.4196	0.4308
Case2020	CenterB	pure_edema	nnunet_oof	0.5631	0.7019

附录 E2: case oracle 和 voxel oracle 分块 (续 7)

case_id	center	metric_name	best_model	case_oracle	voxel_oracle
Case2020	CenterB	lesion_union	nnunet_oof	0.7695	0.7923
Case2031	CenterB	scar	nnunet_oof	0.8014	0.8973
Case2031	CenterB	pure_edema	nnunet_oof	0.2988	0.7984
Case2031	CenterB	lesion_union	nnunet_oof	0.8013	0.9197
Case2033	CenterB	scar	mosaic_oof	0.7441	0.9777
Case2033	CenterB	pure_edema	nnunet_oof	0.523	0.9168
Case2033	CenterB	lesion_union	nnunet_oof	0.7858	0.9875
Case3004	CenterC	scar	nnunet_oof	0.6247	0.8873

附录 E2: case oracle 和 voxel oracle 分块 (续 8)

case_id	center	metric_name	best_model	case_oracle	voxel_oracle
Case3004	CenterC	pure_edema	nnunet_oof	0.453	0.8063
Case3004	CenterC	lesion_union	nnunet_oof	0.7494	0.9806
Case3011	CenterC	scar	mosaic_oof	0.7138	0.8495
Case3011	CenterC	pure_edema	nnunet_oof	0.2667	0.5665
Case3011	CenterC	lesion_union	mosaic_oof	0.7541	0.8861
Case3012	CenterC	scar	nnunet_oof	0.8267	0.8777
Case3012	CenterC	pure_edema	nnunet_oof	0.4507	0.5211
Case3012	CenterC	lesion_union	nnunet_oof	0.7032	0.7578

附录 E2: case oracle 和 voxel oracle 分块 (续 9)

case_id	center	metric_name	best_model	case_oracle	voxel_oracle
Case3023	CenterC	scar	mosaic_oof	0.663	0.9762
Case3023	CenterC	pure_edema	mosaic_oof	0.3463	0.5878
Case3023	CenterC	lesion_union	nnunet_oof	0.7186	0.9637
Case3026	CenterC	scar	nnunet_oof	0.8082	0.9205
Case3026	CenterC	pure_edema	nnunet_oof	0.2192	0.4874
Case3026	CenterC	lesion_union	mosaic_oof	0.7943	0.9612
Case3034	CenterC	scar	nnunet_oof	0.8294	0.9993
Case3034	CenterC	pure_edema	nnunet_oof	0.6546	0.8522

附录 E2: case oracle 和 voxel oracle 分块 (续 10)

case_id	center	metric_name	best_model	case_oracle	voxel_oracle
Case3034	CenterC	lesion_union	nnunet_oof	0.82	0.9771
Case3038	CenterC	scar	nnunet_oof	0.7583	0.8583
Case3038	CenterC	pure_edema	nnunet_oof	0.1678	0.5392
Case3038	CenterC	lesion_union	nnunet_oof	0.7429	0.9254
Case3040	CenterC	scar	mosaic_oof	0.8966	0.9405
Case3040	CenterC	pure_edema	nnunet_oof	0.2491	0.9073
Case3040	CenterC	lesion_union	nnunet_oof	0.8685	0.9888
Case3044	CenterC	scar	nnunet_oof	0.7845	0.8407

附录 E2: case oracle 和 voxel oracle 分块 (续 11)

case_id	center	metric_name	best_model	case_oracle	voxel_oracle
Case3044	CenterC	pure_edema	nnunet_oof	0.1661	0.849
Case3044	CenterC	lesion_union	mosaic_oof	0.8614	0.9558
Case5005	CenterE	scar	nnunet_oof	0.5726	0.6074
Case5005	CenterE	lesion_union	nnunet_oof	0.5726	0.6373
Case6001	CenterF	scar	nnunet_oof	0.446	0.7146
Case6001	CenterF	lesion_union	nnunet_oof	0.446	0.7446
Case6010	CenterF	scar	nnunet_oof	0.4926	0.5942
Case6010	CenterF	lesion_union	nnunet_oof	0.4926	0.8541

附录 E2: case oracle 和 voxel oracle 分块 (续 12)

case_id	center	metric_name	best_model	case_oracle	voxel_oracle
Case7005	CenterG	scar	nnunet_oof	0	1
Case7005	CenterG	lesion_union	nnunet_oof	0	1
Case8003	CenterH	scar	mosaic_oof	0.7372	0.8895
Case8003	CenterH	lesion_union	nnunet_oof	0.7173	0.9876
Case8011	CenterH	scar	mosaic_oof	0.2871	0.6881
Case8011	CenterH	lesion_union	nnunet_oof	0.2241	0.8529
Case8015	CenterH	scar	nnunet_oof	0.4693	0.6083
Case8015	CenterH	lesion_union	nnunet_oof	0.4693	0.9469

附录 E3: PRISM 13 checkpoint corrected metrics 分块

step	case_id	metric_name	dice	nnunet	delta
500	Case1004	scar	0.5301	0.8826	-0.3525
500	Case1004	pure_edema	1	1	0
500	Case1004	edema_zone	0	0.8826	-0.8826
500	Case1008	scar	0.3511	0.8739	-0.5228
500	Case1008	pure_edema	1	1	0
500	Case1008	edema_zone	0	0.8739	-0.8739
500	Case1020	scar	0.3019	0.7594	-0.4575
500	Case1020	pure_edema	1	1	0

附录 E3: PRISM 13 checkpoint corrected metrics 分块 (续 2)

step	case_id	metric_name	dice	nnunet	delta
500	Case1020	edema_zone	0	0.7594	-0.7594
500	Case1026	scar	0.3089	0.8388	-0.5299
500	Case1026	pure_edema	1	1	0
500	Case1026	edema_zone	0	0.8388	-0.8388
500	Case1032	scar	0.3139	0.7676	-0.4537
500	Case1032	pure_edema	1	1	0
500	Case1032	edema_zone	0	0.7676	-0.7676
500	Case1035	scar	0.3732	0.875	-0.5018

附录 E3: PRISM 13 checkpoint corrected metrics 分块 (续 3)

step	case_id	metric_name	dice	nnunet	delta
500	Case1035	pure_edema	1	1	0
500	Case1035	edema_zone	0	0.875	-0.875
500	Case1039	scar	0.3547	0.8657	-0.5111
500	Case1039	pure_edema	1	1	0
500	Case1039	edema_zone	0	0.8657	-0.8657
500	Case1043	scar	0.3403	0.7731	-0.4327
500	Case1043	pure_edema	1	1	0
500	Case1043	edema_zone	0	0.7731	-0.7731

附录 E3: PRISM 13 checkpoint corrected metrics 分块 (续 4)

step	case_id	metric_name	dice	nnunet	delta
500	Case1056	scar	0.2977	0.8541	-0.5564
500	Case1056	pure_edema	1	1	0
500	Case1056	edema_zone	0	0.8541	-0.8541
500	Case1058	scar	0.4004	0.9069	-0.5065
500	Case1058	pure_edema	1	1	0
500	Case1058	edema_zone	0	0.9069	-0.9069
500	Case1071	scar	0.2651	0.8407	-0.5756
500	Case1071	pure_edema	1	1	0

附录 E3: PRISM 13 checkpoint corrected metrics 分块 (续 5)

step	case_id	metric_name	dice	nnunet	delta
500	Case1071	edema_zone	0	0.8407	-0.8407
500	Case1074	scar	0.2748	0.9189	-0.6441
500	Case1074	pure_edema	1	1	0
500	Case1074	edema_zone	0	0.9189	-0.9189
500	Case1077	scar	0.1608	0.8127	-0.6519
500	Case1077	pure_edema	1	1	0
500	Case1077	edema_zone	0	0.8127	-0.8127
500	Case2009	scar	0.2979	0.5996	-0.3017

附录 E3: PRISM 13 checkpoint corrected metrics 分块 (续 6)

step	case_id	metric_name	dice	nnunet	delta
500	Case2009	pure_edema	0.2362	0.8094	-0.5731
500	Case2009	edema_zone	0.5784	0.7874	-0.2089
500	Case2011	scar	0.4509	0.9236	-0.4727
500	Case2011	pure_edema	0.2545	0.8881	-0.6336
500	Case2011	edema_zone	0.7252	0.9363	-0.2111
500	Case2012	scar	0.0883	0.6156	-0.5274
500	Case2012	pure_edema	0.2305	0.7723	-0.5418
500	Case2012	edema_zone	0.2376	0.7309	-0.4933

附录 E3: PRISM 13 checkpoint corrected metrics 分块 (续 7)

step	case_id	metric_name	dice	nnunet	delta
500	Case2016	scar	0.3592	0.8874	-0.5283
500	Case2016	pure_edema	0.2171	0.7967	-0.5796
500	Case2016	edema_zone	0.5742	0.8869	-0.3127
500	Case2018	scar	0.1959	0.9197	-0.7238
500	Case2018	pure_edema	0.1377	0.8875	-0.7498
500	Case2018	edema_zone	0.4503	0.939	-0.4887
500	Case2019	scar	0.3024	0.9268	-0.6244
500	Case2019	pure_edema	0.168	0.8462	-0.6782

附录 E3: PRISM 13 checkpoint corrected metrics 分块 (续 8)

step	case_id	metric_name	dice	nnunet	delta
500	Case2019	edema_zone	0.6411	0.8901	-0.249
500	Case2021	scar	0.5314	0.9027	-0.3713
500	Case2021	pure_edema	0.1847	0.845	-0.6603
500	Case2021	edema_zone	0.5959	0.9286	-0.3327
500	Case2023	scar	0.4133	0.8803	-0.467
500	Case2023	pure_edema	0.3558	0.796	-0.4402
500	Case2023	edema_zone	0.6405	0.8879	-0.2473
500	Case2034	scar	0.3837	0.8926	-0.5089

附录 E3: PRISM 13 checkpoint corrected metrics 分块 (续 9)

step	case_id	metric_name	dice	nnunet	delta
500	Case2034	pure_edema	0.2193	0.8288	-0.6095
500	Case2034	edema_zone	0.4834	0.8943	-0.411
500	Case3009	scar	0.2312	0.9245	-0.6933
500	Case3009	pure_edema	0.0878	0.8288	-0.741
500	Case3009	edema_zone	0.2231	0.9179	-0.6948
500	Case3014	scar	0.4738	0.8971	-0.4233
500	Case3014	pure_edema	0.1475	0.7489	-0.6014
500	Case3014	edema_zone	0.6091	0.9073	-0.2982

附录 E3: PRISM 13 checkpoint corrected metrics 分块 (续 10)

step	case_id	metric_name	dice	nnunet	delta
500	Case3017	scar	0.56	0.8898	-0.3298
500	Case3017	pure_edema	0.0858	0.794	-0.7082
500	Case3017	edema_zone	0.551	0.9045	-0.3535
500	Case3028	scar	0.3645	0.8845	-0.52
500	Case3028	pure_edema	0.2745	0.9186	-0.6441
500	Case3028	edema_zone	0.4914	0.919	-0.4277
500	Case3032	scar	0	0	0
500	Case3032	pure_edema	0.0616	0.9619	-0.9003

附录 E3: PRISM 13 checkpoint corrected metrics 分块 (续 11)

step	case_id	metric_name	dice	nnunet	delta
500	Case3032	edema_zone	0.4835	0.9558	-0.4724
500	Case3036	scar	0.6897	0.909	-0.2193
500	Case3036	pure_edema	0.1671	0.8001	-0.633
500	Case3036	edema_zone	0.643	0.8957	-0.2527
500	Case3042	scar	0.5471	0.9563	-0.4092
500	Case3042	pure_edema	0.0703	0.8499	-0.7796
500	Case3042	edema_zone	0.6938	0.9606	-0.2668
500	Case7006	scar	0	1	-1

附录 E3: PRISM 13 checkpoint corrected metrics 分块 (续 12)

step	case_id	metric_name	dice	nnunet	delta
500	Case7006	pure_edema	1	1	0
500	Case7006	edema_zone	1	1	0
500	Case8009	scar	0.2169	0.8006	-0.5838
500	Case8009	pure_edema	1	1	0
500	Case8009	edema_zone	0	0.8006	-0.8006
500	Case8012	scar	0.1249	0.8958	-0.7709
500	Case8012	pure_edema	1	1	0
500	Case8012	edema_zone	0	0.8958	-0.8958

附录 E4: Batch0-7 / SRR casewise metrics 分块

case_id	pathology	anchor	srr_dice	dice_delta	srr_hd95
Case1002	edema				0
Case1002	scar	0.6167	0.6167	0	4.3235
Case1007	edema				0
Case1007	scar	0.5811	0.5811	0	6.836
Case1009	edema				0
Case1009	scar	0.6049	0.6049	0	8.4968
Case1010	edema				0
Case1010	scar	0.4828	0.4828	0	9.1033

附录 E4: Batch0-7 / SRR casewise metrics 分块 (续 2)

case_id	pathology	anchor	srr_dice	dice_delta	srr_hd95
Case1021	edema				0
Case1021	scar	0.5699	0.5699	0	7.9721
Case1023	edema				0
Case1023	scar	0.6466	0.6466	0	5
Case1029	edema				0
Case1029	scar	0.1644	0.1644	0	51.3953
Case1033	edema				0
Case1033	scar	0.6714	0.6714	0	7.152

附录 E4: Batch0-7 / SRR casewise metrics 分块 (续 3)

case_id	pathology	anchor	srr_dice	dice_delta	srr_hd95
Case1040	edema				0
Case1040	scar	0.428	0.428	0	8.5
Case1042	edema				0
Case1042	scar	0.7574	0.7574	0	3.0572
Case1045	edema				0
Case1045	scar	0.0997	0.0997	0	58.2394
Case1047	edema				0
Case1047	scar	0.6283	0.6283	0	5.7812

附录 E4: Batch0-7 / SRR casewise metrics 分块 (续 4)

case_id	pathology	anchor	srr_dice	dice_delta	srr_hd95
Case1053	edema				0
Case1053	scar	0.5196	0.5196	0	27.5368
Case1062	edema				0
Case1062	scar	0.5071	0.5071	0	15.2911
Case1070	edema				0
Case1070	scar	0.5005	0.5005	0	34.2974
Case1073	edema				0
Case1073	scar	0.4283	0.4283	0	10.6782

附录 E4: Batch0-7 / SRR casewise metrics 分块 (续 5)

case_id	pathology	anchor	srr_dice	dice_delta	srr_hd95
Case1080	edema				0
Case1080	scar	0.6369	0.6369	0	11
Case2002	edema	0.5376	0.5376	0	17.4532
Case2002	scar	0.5603	0.5603	0	21.4154
Case2007	edema	0.5701	0.5701	0	7.513
Case2007	scar	0.4899	0.4899	0	11.683
Case2008	edema	0.3486	0.3486	0	33.2379
Case2008	scar	0.7532	0.7532	0	6.6406

附录 E4: Batch0-7 / SRR casewise metrics 分块 (续 6)

case_id	pathology	anchor	srr_dice	dice_delta	srr_hd95
Case2017	edema	0.6796	0.6793	-0.0003	10
Case2017	scar	0.5471	0.5471	0	11.3389
Case2020	edema	0.5631	0.5631	0	7.513
Case2020	scar	0.4196	0.4196	0	20.9944
Case2031	edema	0.2988	0.2988	0	25.053
Case2031	scar	0.8014	0.8014	0	5.0087
Case2033	edema	0.523	0.5228	-0.0002	10.7645
Case2033	scar	0.7209	0.7209	0	10

附录 E4: Batch0-7 / SRR casewise metrics 分块 (续 7)

case_id	pathology	anchor	srr_dice	dice_delta	srr_hd95
Case3004	edema	0.453	0.453	0	9.222
Case3004	scar	0.6247	0.6247	0	24
Case3011	edema	0.2667	0.2667	0	34.2661
Case3011	scar	0.639	0.639	0	10.9603
Case3012	edema	0.4507	0.4506	-0.0001	22.4114
Case3012	scar	0.8267	0.8267	0	2.9163
Case3023	edema	0.1629	0.1629	0	41.3249
Case3023	scar	0.6613	0.6613	0	4.1248

附录 E4: Batch0-7 / SRR casewise metrics 分块 (续 8)

case_id	pathology	anchor	srr_dice	dice_delta	srr_hd95
Case3026	edema	0.2192	0.2192	0	17.3257
Case3026	scar	0.8082	0.8082	0	2.9163
Case3034	edema	0.6546	0.6544	-0.0002	9.1352
Case3034	scar	0.8294	0.8294	0	1.6302
Case3038	edema	0.1678	0.1678	0	28.2129
Case3038	scar	0.7583	0.7583	0	8.2979
Case3040	edema	0.2491	0.2491	0	24.1389
Case3040	scar	0.8691	0.8691	0	1.4581

附录 E4: Batch0-7 / SRR casewise metrics 分块 (续 9)

case_id	pathology	anchor	srr_dice	dice_delta	srr_hd95
Case3044	edema	0.1661	0.1661	0	22.6122
Case3044	scar	0.7845	0.7845	0	4.8907
Case5005	edema				0
Case5005	scar	0.5726	0.5726	0	11.149
Case6001	edema				0
Case6001	scar	0.446	0.446	0	10.6066
Case6010	edema				0
Case6010	scar	0.4926	0.4926	0	13.0504

附录 E4: Batch0-7 / SRR casewise metrics 分块 (续 10)

case_id	pathology	anchor	srr_dice	dice_delta	srr_hd95
Case7005	edema				0
Case7005	scar	0	0	0	
Case8003	edema				0
Case8003	scar	0.7173	0.7173	0	8.7543
Case8011	edema				0
Case8011	scar	0.2241	0.2241	0	31.3276
Case8015	edema				0
Case8015	scar	0.4693	0.4693	0	20.0467

附录 E4: Batch0-7 / SRR casewise metrics 分块 (续 11)

case_id	pathology	anchor	srr_dice	dice_delta	srr_hd95
Case8019	edema				0
Case8019	scar	0.6588	0.6588	0	11.25
Case8021	edema				0
Case8021	scar	0	0	0	
Case8022	edema				0
Case8022	scar	0.6749	0.6749	0	10.093
Case8023	edema				0
Case8023	scar	0.4551	0.4551	0	32.0091

附录 E5: ARC casewise metrics 分块

case_id	variant	pathology	dice	hd95	mask_delta
Case1002	raw_direct	scar	0.3812	15.3162	0.759
Case1002	raw_direct	edema_zone	0.3812	15.3162	0.759
Case1002	raw_direct	pure_edema		0	0
Case1002	postproc	scar	0.4098	16.0899	0.8099
Case1002	postproc	edema_zone	0.4098	16.0899	0.8099
Case1002	postproc	pure_edema		0	0
Case1002	nnunet	scar	0.6167	4.3235	0
Case1002	nnunet	edema_zone	0.6167	4.3235	0

附录 E5: ARC casewise metrics 分块 (续 2)

case_id	variant	pathology	dice	hd95	mask_delta
Case1002	nnunet	pure_edema		0	0
Case1002	raw_identity	scar	0.3812	15.3162	0.759
Case1002	raw_identity	edema_zone	0.3812	15.3162	0.759
Case1002	raw_identity	pure_edema		0	0
Case1002	postprocessed...	scar	0.4098	16.0899	0.8099
Case1002	postprocessed...	edema_zone	0.4098	16.0899	0.8099
Case1002	postprocessed...	pure_edema		0	0
Case1007	raw_direct	scar	0.5336	5.89	1.1359

附录 E5: ARC casewise metrics 分块 (续 3)

case_id	variant	pathology	dice	hd95	mask_delta
Case1007	raw_direct	edema_zone	0.5336	5.89	1.1359
Case1007	raw_direct	pure_edema		0	0
Case1007	postproc	scar	0.529	6.3482	1.1993
Case1007	postproc	edema_zone	0.529	6.3482	1.1993
Case1007	postproc	pure_edema		0	0
Case1007	nnunet	scar	0.5811	6.836	0
Case1007	nnunet	edema_zone	0.5811	6.836	0
Case1007	nnunet	pure_edema		0	0

附录 E5: ARC casewise metrics 分块 (续 4)

case_id	variant	pathology	dice	hd95	mask_delta
Case1007	raw_identity	scar	0.5336	5.89	1.1359
Case1007	raw_identity	edema_zone	0.5336	5.89	1.1359
Case1007	raw_identity	pure_edema		0	0
Case1007	postprocessed...	scar	0.529	6.3482	1.1993
Case1007	postprocessed...	edema_zone	0.529	6.3482	1.1993
Case1007	postprocessed...	pure_edema		0	0
Case1009	raw_direct	scar	0.5537	10.8904	0.4993
Case1009	raw_direct	edema_zone	0.5537	10.8904	0.4993

附录 E5: ARC casewise metrics 分块 (续 5)

case_id	variant	pathology	dice	hd95	mask_delta
Case1009	raw_direct	pure_edema		0	0
Case1009	postproc	scar	0.5547	10.884	0.5232
Case1009	postproc	edema_zone	0.5547	10.884	0.5232
Case1009	postproc	pure_edema		0	0
Case1009	nnunet	scar	0.6049	8.4968	0
Case1009	nnunet	edema_zone	0.6049	8.4968	0
Case1009	nnunet	pure_edema		0	0
Case1009	raw_identity	scar	0.5537	10.8904	0.4993

附录 E5: ARC casewise metrics 分块 (续 6)

case_id	variant	pathology	dice	hd95	mask_delta
Case1009	raw_identity	edema_zone	0.5537	10.8904	0.4993
Case1009	raw_identity	pure_edema		0	0
Case1009	postprocessed...	scar	0.5547	10.884	0.5232
Case1009	postprocessed...	edema_zone	0.5547	10.884	0.5232
Case1009	postprocessed...	pure_edema		0	0
Case1010	raw_direct	scar	0.3668	14.004	1.1158
Case1010	raw_direct	edema_zone	0.3668	14.004	1.1158
Case1010	raw_direct	pure_edema		0	0

附录 E5: ARC casewise metrics 分块 (续 7)

case_id	variant	pathology	dice	hd95	mask_delta
Case1010	postproc	scar	0.4024	13.9651	1.166
Case1010	postproc	edema_zone	0.4024	13.9651	1.166
Case1010	postproc	pure_edema		0	0
Case1010	nnunet	scar	0.4828	9.1033	0
Case1010	nnunet	edema_zone	0.4828	9.1033	0
Case1010	nnunet	pure_edema		0	0
Case1010	raw_identity	scar	0.3668	14.004	1.1158
Case1010	raw_identity	edema_zone	0.3668	14.004	1.1158

附录 E5: ARC casewise metrics 分块 (续 8)

case_id	variant	pathology	dice	hd95	mask_delta
Case1010	raw_identity	pure_edema		0	0
Case1010	postprocessed...	scar	0.4024	13.9651	1.166
Case1010	postprocessed...	edema_zone	0.4024	13.9651	1.166
Case1010	postprocessed...	pure_edema		0	0
Case1021	raw_direct	scar	0.5234	9.5704	0.5997
Case1021	raw_direct	edema_zone	0.5234	9.5704	0.5997
Case1021	raw_direct	pure_edema		0	0
Case1021	postproc	scar	0.5278	9.9658	0.654

附录 E5: ARC casewise metrics 分块 (续 9)

case_id	variant	pathology	dice	hd95	mask_delta
Case1021	postproc	edema_zone	0.5278	9.9658	0.654
Case1021	postproc	pure_edema		0	0
Case1021	nnunet	scar	0.5699	7.9721	0
Case1021	nnunet	edema_zone	0.5699	7.9721	0
Case1021	nnunet	pure_edema		0	0
Case1021	raw_identity	scar	0.5234	9.5704	0.5997
Case1021	raw_identity	edema_zone	0.5234	9.5704	0.5997
Case1021	raw_identity	pure_edema		0	0

附录 E6：历史 prediction binding 摘要

逐项 prediction 路径很长，完整清单保留在 historical_prediction_binding.csv。PDF 只展示按模型、资产类型和绑定状态聚合的计数，避免路径列覆盖其它列。

model_id	artifact	status	rows
B0_3_SRR_ANCHOR	prediction	BOUND	120
B0_3_SRR_ANCHOR	prediction	BOUND_TRUNCATED_FOR_P...	1
DG_DR_DPR	prediction	BOUND	119

附录 E7：组件生存清单分块

source_model	component	casewise_signal	failure_mode	future
Batch7	availability-awa...	some	implementation c...	KEEP_DATA_RULE
Batch7	pathology-specif...	weak	not deployable	RETEST_OPTION
Batch7	negative-space	unproven	not independentl...	RETEST_OPTION
Batch7	complex router/SIP	harmful	scar degradation	DO_NOT_REUSE_CUR...
MMRD	reliable-label s...	supportive	not model-gain a...	KEEP_DATA_RULE
MMRD	no-T2 edema hygiene	supportive	not model-gain a...	KEEP_DATA_RULE
MMRD	simple residual ...	weak	underpowered head	DO_NOT_REUSE_CUR...
Cascade	strong baseline ...	protective	gain near zero	KEEP_STRONG

附录 E7：组件生存清单分块（续 2）

source_model	component	casewise_signal	failure_mode	future
Cascade	bounded correction	small	ceiling too low	RETEST_OPTION
Cascade	prototype input	unresolved	control shared i...	UNRESOLVED
ARC	direct reconstru...	mixed	decoder/final-ma...	RETEST_OPTION
ARC	decoder reset	harmful	random decoder l...	DO_NOT_REUSE
DG/DR/DPR	pathology-specif...	unresolved	stopped/partial ...	RETEST_OPTION
PRISM	private pyramids...	weak	decoder/training...	DO_NOT_REUSE_CUR...
PRISM	stage schedule	declines late	selected checkpo...	RETEST_OPTION

附录 E：代码、配置、split 和预测 hash

当前 hash manifest 是启动级定位清单。大型 checkpoint 和 prediction 在 V2 中保留 path/size 绑定；关键 source 和小文件保留 SHA。缺 exact replay 条件的模型在对应 binding 表中标注。

path	hash_status	size_bytes
results/.../finalizer_state.json	FULL	1260
results/.../checkpoint_resume...	FULL	594
results/.../controller_context...	FULL	2190
results/.../w3_training_summa...	FULL	2759
results/.../controller_context...	PREFIX_8192_BYTES	8289
results/.../controller_bootst...	FULL	894
results/.../architecture_delt...	FULL	4306
results/.../known_bad_report...	FULL	443
results/.../unit_test_report...	FULL	327
results/.../controller_report.md	FULL	1976
results/.../mapper_report_fin...	PREFIX_8192_BYTES	9056
results/.../adoption_receipt...	FULL	1524

附录 F：证据缺口

V2 的缺口不再是 REQUIRED GPU 未运行，而是后续科学设计前的边界：不能把 oracle 写成可部署性能，不能把 full-data MoSAIC 写成 clean 架构优势，不能复制历史失败实现。

附录 G：PDF 渲染验收记录

最终 PDF 采用 `pandoc_xelatex_named_fonts`，不是 `Chromium fallback`。验收重点是 `pdftotext` 不含 `HeadlessChrome/Skia`，`pdffonts` 出现 `TeXGyreTermes` 与 `/users render bundle` 的 `NotoSerifSC/NotoSansSC`，`pdftotext -layout` 中文可抽取，并完成全页 PNG 视觉检查。